MAKING THE RIGHT CONNECTIONS:
IMPROVING THE COMPREHENSION OF STRUGGLING READERS

GRANT# R324A110046
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I. ACCOMPLISHMENTS

A. Major Goals of the Project

The major goals of this project, as specified in the performance agreement, are to produce one or more fully developed questioning intervention(s) aimed at improving the reading comprehension of students at risk or identified as having disabilities. Intended products include: (a) one or more prototypes that are feasible and promising, with manuals; (b) one or more observational checklists to be used to document fidelity; and (c) training materials to prepare school-based personnel for implementation. We also aim to identify for whom (subgroups of struggling readers) and under what conditions (type and timing of questions, intensity of intervention) the intervention is most promising.

B. What was Accomplished under these Goals in Year 3

In Year 3, we completed or are in process of completing the following activities to meet the above project goals: (a) Finalize core intervention questions, materials, and procedures to be implemented in the Year 3 pilot study (complete); (b) Develop materials and procedures to train school-based personnel to implement intervention (complete); (c) Recruit teachers to implement the intervention (complete); (d) Train the teachers to implement the intervention (complete); (e) Identify struggling readers (complete); (f) Conduct pilot study (complete); (g) Analyze and interpret data from pilot study to examine feasibility and student outcomes (in process); and (h) Finalize the intervention based on results of Year 3 study activities (in process). These activities fall into three main categories: Identifying and Describing Struggling Readers, Text Selection and Development, and Intervention Development. Figure 1 provides details of activities under each category. Specific methods and results for each category are described in detail below.

Figure 1. Year 3 Timeline

<table>
<tr><th></th><th>Identifying and Describing Struggling Readers</th><th>Revising Existing Texts & Questions</th><th>Refining Intervention</th></tr>
<tr><td>Sep-13</td><td>Screening Phase 1</td><td rowspan="10">Revised existing texts and questions based on feedback from Year 2.</td><td>Round 11: In-House Revision of Intervention</td></tr>
<tr><td>Oct-13</td><td rowspan="3">Screening Phase 2 & Think-Aloud Administration</td><td rowspan="2">Round 12: Fall Trial Run and continued revision; Development of teacher training; RLA Observations.</td></tr>
<tr><td>Nov-13</td></tr>
<tr><td>Dec-13</td><td>Think-Aloud Coding</td></tr>
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<tr><td>Jun-14</td><td>Subgroup ID</td><td rowspan="3">Final revision of Texts and Questions</td><td>Spring Pilot recall coding</td></tr>
<tr><td>Jul-14</td><td>Data analysis</td><td>Data analysis</td></tr>
<tr><td>Aug-14</td><td></td><td>Round 15: Final Revision of Intervention</td></tr>
</table>

IDENTIFYING AND DESCRIBING STRUGGLING READERS

The purpose of identifying and describing struggling readers was to (a) identify students struggling primarily with comprehension, and thus eligible for the intervention to be developed and (b) identify subgroups of struggling comprehenders. Specific research questions include:

(1) Of a diverse sample of fourth-grade students, which students are struggling primarily with reading comprehension (not decoding/word recognition or fluency skills)?
(2) Of those students identified as struggling comprehenders, do subgroups of comprehenders emerge, based on responses to a think-aloud task?

Method

Participants

Districts and schools. Approval to conduct this research was obtained from two districts' Departments of Research, Evaluation & Assessment (REA) prior to recruiting teachers and students. District 1 is a large urban district serving about 32,283 K-12 students in 167 schools at the time of this study (2012-13 school year). District 2 is a first-ring suburb of a large urban area serving 10,330 K-12 students in 24 schools at the time of this study. A total of seven schools across the two districts agreed to participate. District and school demographic information for the study year (2013-14) are summarized in Table 1.

Table 1. ***Participating District and School Demographic Information***

	Total Enrollment	Students of Color (%)	English Learners (%)	Free/Reduced Lunch (%)	Special Ed (%)
District 1	36,283	66.9	24.6	64.8	18.2
School 1 (BM)	454	85.5	25.8	86.8	20.3
School 2 (LHU)	639	17.2	3	8	9.1
School 3 (Gr)	558	94.3	63.3	95.5	17
School 4 (Hale)	621	22.7	2.9	15.5	8.5
District 2	10,330	45.6	12.2	41.3	14.5
School 5 (NH)	506	41.7	17.6	38.7	10.7
School 6 (Ol)	529	38.2	14.4	32.7	9.3
School 7 (VVE)	539	87.6	47.5	85	8

At the end of Year 2, we invited all participating teachers to participate in Year 3. Most teachers at least tentatively agreed (except for teachers who knew they were leaving their current position or grade level). In fall of Year 3, we followed up with all teachers. In cases where Year 2 teachers were no longer available, new teachers were brought on board (e.g., in one school, there was a new 4th-grade teacher in an already-participating school; in another school, teachers "loop" such that Year 2 teachers moved on to 5th grade, and Year 1 teachers who had not participated in Year 2 were now available for Year 3). We met with grade-level teams to provide an overview of the study and to address questions. In some cases, reading specialists and/or special education teachers also attended the meetings. We contacted school principals to confirm their approval of the teachers' participation. Table 2 provides a summary of teachers who returned, withdrew, or enrolled as new teachers.

Table 2. ***Summary of New and Participating Teachers***

	Returning	Withdrawn	Newly Enrolled
District 1			
School 1 (BM)	1 gr. 4	1 gr. 4 (moved to new position)	0
School 2 (LHU)	2 gr. 4	2 gr. 4 (no longer teaching reading)	0
School 3 (Gr)	1 gr. 4, 1 ELL specialist, 1 sp. ed. teacher	0	1 new gr. 4 teacher
School 4 (H)	4 gr. 4, 1 sp. ed. teacher	0	0
District 2			
School 5 (NH)	3 gr. 4; 2 reading specialists 1 sp ed teacher	1 (moved)	1
School 6 (Ol)	0	4 (moved to gr. 5)	3 gr. 4 (from Yr. 1)
School 7 (VVE)	3 gr. 4	1 gr. 4; 1 ELL (moved to new schools)	1 new gr. 4 teacher
TOTALS	20	9	6

Teachers. Twenty-six teachers participated, including 20 fourth-grade classroom teachers and 6 specialists (including one ELL specialist, three special education teachers, and two intervention specialists). Teacher-reported demographic data are in Table 3.

Students. All fourth-grade students in participating classrooms were invited to participate in the study. Research staff provided brief (5-10 min) presentations of the study to each participating class, and distributed consent forms to all students to take home and have signed by their parents. Students received gel pens when they returned the signed consent forms (whether the parents indicated 'yes' or 'no' to participation). Table 4 shows the distribution and return rates of parental consent forms in the two districts.

Measures

Screening measures. Screening occurred in two phases. Phase I screening measures included CBM maze tests. Phase II screening measures included CBM oral reading and district tests. Table 5 shows intercorrelations among screening measures.

Curriculum-Based Measurement (CBM)—Maze Task. The CBM maze task (e.g., Fuchs & Fuchs, 1992) is designed to be a brief assessment of overall reading proficiency. The maze is group administered, and takes about 10-15 min to administer. In this study, students read two passages in which every seventh word was deleted and replaced with three word choices, only one of which fit in the sentence. Students completed two practice sentences, and then read each passage silently for 2.5 min, selecting the words that best fit in the sentences. The score was the average correct maze selections made on the two passages. Alternate-form reliability coefficients have been reported to range from r = .83 to .98, and correlations between maze selection scores and other reading measures typically range from r = .60 to .80 for elementary-aged students (Wayman, Wallace, Wiley, Ticha, & Espin, 2007). In the screening sample for this study (n = 460), test-retest reliability for the maze was r = .89, p < .001.

Table 3. ***Teacher-Reported Demographics***

	Classroom teachers (*n*=20)		Specialists (*n*=6)	
	n	%	n	%
Sex (female)	15	75	6	100
Race				
White	18	90	6	100
Hispanic	1	5	0	0
White, Asian, American Indian	1	5	0	0
Age				
20-29	0	0	1	17
30-39	8	40	1	17
40-49	6	30	2	33
50-59	5	25	1	17
60 +	1	5	1	17
Highest Degree				
Bachelor's	3	15	2	33
Master's	8	40	0	0
Master's + Additional Coursework	9	45	4	67
	M (range)	SD	*M* (range)	SD
Years in current position	8.80 (1-31)	8.94	7.83 (2-25)	9.00
Years in general education	17.25 (3-36)	8.87	8.50 (0-18)	8.67
Years teaching Grade 4	7.80 (1-22)	7.00	6.83 (0-20)	8.01
Years teaching special ed.	1.20 (0-13)	3.56	6.67 (0-25)	10.31
Years as a reading specialist	0	0.00	.40 (0-2)	0.89
Hours of PD in reading comprehension	9.95 (0-32)	8.65	7.50 (4-11)	4.95
Students in class/caseload	25.50 (19-33)	4.79	29.17 (18-40)	10.61

Table 4. ***Distribution and Return Rates of Consent Forms***

	Distributed	Returned Indicating 'Yes' or 'No' to Participation			Return Rate
		Yes	No	Total	
District 1 (MPS)	366	286	37	323	88%
District 2 (BPS)	238	175	30	205	86%
Total	604	461	67	528	87%

District tests. Both districts used the Measures of Academic Progress (MAP), developed by the Northwest Evaluation Association to assess students' reading achievement and growth from late elementary through high school. Reading achievement is assessed in Minnesota in the

areas of "Literature," "Informational Text," "Foundational Skills and Vocabulary" and "Language Use and Structure." These areas are aligned with district and state reading standards. All MAP assessments are computerized and presented adaptively, adjusting in difficulty based on each student's responses to test questions. They commonly consist of more than 40 questions. Assessments are provided in a survey structure with content area goal scores supplementing an overall score. For the purposes of this study, we recorded the Normal Curve Equivalent (NCE) and RIT score, which allows the values of achievement levels and the values of items to be on the same scale. The reported stratified, randomly parallel form reliability for 4th grade is .80 (Northwest Evaluation Association, 2011).

CBM oral reading. The CBM oral reading task (Deno, 1985) consists of individually administered grade-level passages, which students read aloud for 1 min. As the student reads, the examiner marks insertions, omissions, substitutions, and mispronunciations not caused by speech-related problems or dialects. Errors are not corrected and no decoding assistance is given. In this study, the student read two passages; the score is the mean number of words read correctly in one min. For this study, we used two Formative Assessment System for Teachers (FAST; Christ, Ardoin, Eckert, & White, 2010a) 4th-grade-level CBM passages. Criterion validity for the 4th-grade passages with other reading measures (DIBELS and TOSREC) has been reported as r = .95 (DIBELS) and .74 (TOSREC), and classification accuracy for identifying students at risk has been reported as strong (Christ, Ardoin, Eckert, & White, 2010b). In the screening sample for this study (n = 142), test-retest reliability was r = .90, p < .001.

Table 5. ***Intercorrelations among Screening Measures***

	1. Maze	2. MAP	3. CBM Oral Reading
1. Maze			
2. MAP	.74* (n = 449)		
3. CBM Oral Reading	.72* (n = 142)	.69* (n =138)	.

Note. * Significant correlation coefficients (p < .001). MAP = Measures of Academic Progress. *N*s vary because MAP data were not available for all students, and CBM Oral Reading data were collected only from students in Screening Phase II.

Think-aloud task. The think-aloud task was used to identify possible subgroups of struggling comprehenders. As in previous related studies (e.g., McMaster et al., 2012; Rapp et al., 2007), we used a think-aloud task adapted from Ericsson & Simon (1993). We administered a practice text ("Jimmy and the New Bike") adapted from Magliano, Trabasso, & Graesser, 1999, and two texts, one narrative ("Brian's Magical Skill") and one informational ("Living Underground: The Life of a Mole") adapted from van den Broek et al., 2001). Each sentence from each text was printed on 5" X 6" cards. The two think-aloud texts used in Year 3 are provided in Appendix A.

While four texts were used for the think-aloud procedure in years 1 and 2, the demand on the student and labor-intensity of transcribing and coding inclined us toward administering two rather than four texts, if we could still obtain sufficient results. A comparison of the clusters identified with years 1 and 2 think-alouds indicated a greater overlap between "Brian's Magical Skills" and the narrative texts as a whole (average 93.3%), than the overlap between the second narrative text used previously and the narrative texts (91.8%). There was also a greater overlap in the clusters identified with "Living Underground" and the informational texts as a whole (average 95.0%), than for the second informational text used previously with the informational

texts (average 66.2%). Consequently, we felt confident in administering only two texts (one narrative, one informational) in the Year 3 think-aloud task.

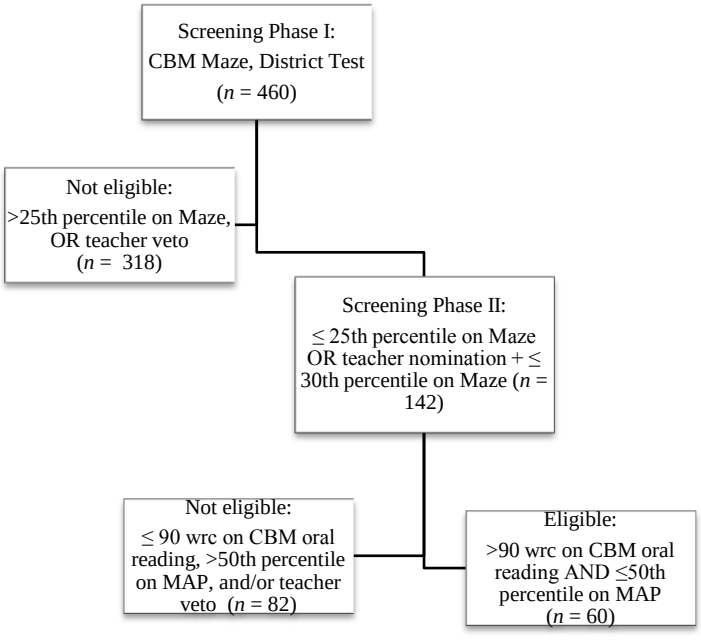
When administering the think-aloud task, students read each text one sentence at a time from cards. The examiner demonstrated the task with the first half of the practice text by giving specific examples of all possible types of responses (e.g., elaborative inferences, paraphrases, predictions). Then, the student practiced thinking aloud with the remaining sentences in the practice text telling the examiner what he or she was thinking. After each text, the examiner asked the student two yes/no questions about the text to check for general comprehension. The student received no decoding help, but received non-leading prompts such as, "What are you thinking now?" if he or she did not think aloud.

Procedures

Screening. To determine eligibility for the intervention, all consented students (n = 460) were screened in two phases (see Figure 2). In Screening Phase I, conducted in September 2013, research staff administered two CBM maze probes. Students whose average scores from the two CBM maze probes were at or below the 25th percentile of the sample were identified as potentially eligible for the study and proceeded to Screening Phase II. The bottom 25th percentile of the two districts together was 15 correct maze choices (cmc) (District 1 = 15, District 2 = 14); we used the full sample percentile.

Those students identified as eligible for Screening Phase II (n = 142), conducted in October-November, 2013, were individually administered two CBM oral reading probes. Those whose mean score was less than or equal to 90 words read correctly (wrc) in one min, or whose teachers felt they would not benefit from intervention due to poor decoding or English language skills, were identified as ineligible for intervention (n = 83), because their reading difficulties were likely at least partly due to decoding or word recognition problems (Christ et al., 2010b). Those who scored greater than 90 wrc in one min (but did not score above the 50th percentile on the district MAP test) were identified as eligible for the intervention (n = 60). Later, one school determined that a child initially identified as eligible for intervention was actually still struggling with decoding and fluency, and withdrew her from the intervention. Thus, intervention data were collected for a total n = 59 students.

Figure 2. ***Participant Flow through Screening Process***



Think-aloud administration and coding. Students identified as eligible for the intervention based on CBM oral reading scores were also administered think-aloud tasks to identify subgroups of struggling readers. The think-aloud task was conducted using a script within the "CBM Oral Reading and Think-Aloud" directions. The Discourse Project Coordinator trained GRAs to administer the think-aloud task by: (1) modeling the procedure, (2) observing each GRA walk through the procedure according to the script, using the story cards, and audio-recording the session, and (3) verifying proficiency of administration by the GRA after a period of independent practice.

Think-aloud tasks were administered individually during Screening Phase II, using one narrative text and one informational text. GRAs were instructed to only administer the think-aloud task if the student had scored at least 90 words read correctly on at least one of the CBM oral reading probes. The entire session was audio-recorded and transcribed.
Undergraduate Research Assistants (URAs) and GRAs transcribed verbatim the think-aloud audio-recordings according to a written procedure. The Discourse Project Coordinator rechecked one transcript made by each URA transcriber, and all transcripts were verified and revised as needed by a second URA transcriber.

Think-aloud transcripts were then parsed into thought units, which represent a complete idea with subject, verb, and possibly subordinate clause. Each parsed thought unit was coded based on a scheme adapted from previous research (Linderholm & van den Broek, 2002) to identify the cognitive processes in which readers engaged during reading. The primary variables coded included:

- Elaborative inferences (readers' attempts to explain the contents of the current sentence using background knowledge. Coded as valid when explanations were constrained by the text, plausible and accurate, or invalid when explanations were not constrained by the text, were implausible or reflected a misunderstanding or other error)

- Predictive inferences (readers' anticipations of what would occur next in the text, also coded as valid or invalid based on whether the predictions were constrained by the text, were plausible or involved error)
- Connecting inferences (readers' explanations of the current sentence by connecting it to the immediately preceding sentence)
- Reinstatement inferences (readers' explanations of the current sentence on the basis of prior text information earlier than the immediately preceding sentence)
- Associations (readers' comments about the current sentence that draw upon background knowledge as an aside)
- Paraphrases (readers' rephrasing of the current sentence in their own words)
- Text repetitions (readers' verbatim reiterations of the current sentence)
- Metacognitive comments (readers' reflections on their understanding of text.)
- Evaluative comments (readers' opinions about the text.)
- Affective responses (readers' reflections of their own emotions in response to the current sentence)
- Questions (readers' questions asked about the content of the current sentence).
- Other (readers' statements that cannot be categorized often because of inaudible utterances)

The coding scheme used in Year 3 reflects the combination of infrequent subcodes that were collapsed during analysis in Year 1. That is, Metacognitive Comment encompassed all types of comments including ones that reflect agreement, disagreement and lack of background knowledge; and Evaluative Comments encompassed evaluations of both content and function of text. During analysis, low frequency codes were collapsed (Paraphrases and text repetitions were combined into one "text paraphrase" code; metacognitive, evaluative, and affective responses were combined into one "comment" code). "Other" codes were excluded from analysis. Frequencies of each code on narrative, informational and all texts will be summarized when analyses are complete (analyses have been delayed due to the need to recode protocols, as described below).

URAs coded the transcribed think-alouds. Training and practice proceeded as follows:

1. ***Training sessions***. The Discourse Project Coordinator and advanced GRA taught the Year 3 URAs the purpose of the think-aloud measure, definitions of codes, and procedures for parsing and coding. They modeled coding and guided practice in coding with protocols from Year 2.
2. ***Coding practice***. Five protocols from Year 2 were distributed for coding practice at lab meetings or in small groups with the project coordinator or advanced GRA. The last narrative and informational texts were coded independently by four pairs of URAs and pairwise consensus calculated. Three pairs were over 80% in agreement, one pair only 70%. The coding practice continued even as the Year 3 think-aloud audiorecordings were being transcribed.
3. ***Paired coding practice***. Pairs of URAs coded transcripts together until they were comfortable with their level of agreement. Then they coded the same protocol independently before reaching consensus on codes for that protocol. When pair-wise IRA on the consensus protocol reached 75%, the pair proceeded to individual coding.
4. ***Individual coding with pairwise consensus coding***. Pairs coded transcripts in sets of five; three each were coded individually, and one in-common transcript was reviewed for

consensus. If pair-wise IRA on the consensus protocol fell below 75%, the pair returned to partner coding. The pairwise agreement ranged from 84-92% (average 87%).

5. ***Third party IRA for coders***. The Discourse Project Coordinator or advanced GRA coded individually-coded protocols and calculated 3rd party IRA. Five URAs had 3rd party IRA below 75%. All protocols coded individuals with low 3rd party IRA and by pairs of individuals who both had low 3rd party IRA were recoded by the project coordinator or advanced GRA. Eighteen protocols were recoded.

The discourse project coordinator and advanced GRA coded protocols until their pairwise IRA exceeded 75%, and then they established IRA with individual URAs. Kappa was calculated between discourse project coordinator or advanced GRA and individual URAs and between pairs of URAs. Kappa ranged from .21 to .91. The URAs with Kappa below .70 were the ones whose individual protocols were recoded.

Results

Research Question 1: Of a diverse sample of fourth-grade students, which students are struggling primarily with reading comprehension (not decoding/word recognition or fluency skills)?

This question was addressed through the screening process described above. Table 6 shows descriptive data obtained from Screening Phases I and II as well as for students identified as eligible for intervention.

Table 6. ***Descriptive Statistics from Screening***

	Screening Phases I and II				
	n	Mean	SD	Min	Max
Maze					
Phase I	460	19.40	7.40	0	44
Phase II	142	11.52	3.78	0	18
	Eligible for Intervention				
	n	Mean	SD	Min	Max
Maze (cmc)	59	11.95	2.45	6	17
MAP SS		187.86	8.69	165	199
MAP Percentile		24.37	14.56	1	49
Oral Reading		114.25	16.86	92	154

Research Question 2: Of those students identified as struggling comprehenders, do subgroups of comprehenders emerge, based on responses to a think-aloud task?

To identify possible subgroups of struggling readers, think-aloud data will be submitted to a cluster analysis using the k-means cluster approach for both the narrative and informational texts. Given that in previous studies, a two-group solution best fit the data, we will use a two-group solution, but will explore a 3-group solution as well to determine if that provides a better fit to the data.

REVISING EXISTING TEXTS AND QUESTIONS

The purpose of revising texts and questions was to refine our pool of stories and informational texts that fit with the schools' instructional programs, were at an appropriate level for the struggling readers in the study, and were amenable to the causal questioning approach under investigation; and to refine causal questions to be used during intervention.

Method

Year 3 texts included the stories and biographies used in Year 2. Existing questions were revised/refined, and new questions developed, with feedback from co-PI van den Broek.

Measures

Coh-Metrix Text Easability Assessor. As in Years 1 and 2, we used the web-based Coh-Metrix Text Easability Assessor (http://141.225.42.101/cohmetrixgates/) to establish the readability as well as other important dimensions of each text. The Text Easability Assessor provides percentile scores on five characteristics of text extracted from a wider range of linguistic features calculated by Coh-Metrix: Narrativity, Syntactic Simplicity, Word Concreteness, Referential Cohesion, and Deep Cohesion. The five text easability scores were extracted from a wide range of linguistic features calculated by Coh-Metrix, and are recommended for use given that they have been found to account for most of the variance in text across grade levels and text types (Graesser, McNamara, & Kulikowich, 2011). Graesser et al. (2011) define these features as follows:

- ***Narrativity***: "The extent to which the text conveys a story, a procedure, or a sequence of episodes of actions and events with animate beings" (p. 228).
- ***Syntactic simplicity***: "Scores are higher when sentences have fewer words and simpler, more familiar syntactic structures" (p. 230).
- ***Word concreteness***: "Scores are higher when a higher percentage of content words are concrete, are meaningful, and evoke mental images—as opposed to being abstract" (p. 230).
- ***Referential cohesion***: "The extent to which explicit content words and ideas in the text are connected with each other as the text unfolds" (p. 230).
- ***Causal cohesion***: "Scores are higher to the extent that clauses and sentences in the text are linked with causal and intentional (goal-oriented) connectives" (p. 230).

Procedures

Text selection. Texts selected for the intervention met the following criteria:

(1) They fit with the types of texts used in participating schools' instructional programs in terms of difficulty level, genre, content, age-appropriateness, and interest level (but did not duplicate specific texts used);
(2) They had to be sufficiently challenging but not too difficult for students struggling primarily with reading comprehension (generally, in the O-R range as identified by teachers in Year 1 of the project);
(3) They had to be short enough to be read in 3 approximately 30 minute sessions; and
(4) They had to have a structure that lent itself to the causal questioning approach to be used in the intervention (e.g., a clear goal with resolution).

Narrative fiction (stories). The following stories were identified for use in the final version of the intervention: "A Different Aladdin," "Little Singing Frog," and "The Emperor and the Peasant Boy." These texts are somewhat similar in genre (they are derivatives or adaptations of fables and folk tales) but very different in content. Each text was divided into three segments to be delivered across three intervention sessions.

Narrative nonfiction (biographies). The following biographies were identified for use in the final version of the intervention: "Jane Goodall and the Chimpanzees," "Cesar Chavez," and "Olympic Dreams." These texts are somewhat similar in genre (they are all biographical) but very different in content (one about a scientist, one about a social activist, and one a collection of three short biographies of Olympic athletes). All texts were determined by reading specialists and discourse experts to be appropriate in terms of difficulty level, genre, content, age-appropriateness, and interest level for struggling fourth-grade readers. Each text was divided into three segments to be delivered across three intervention sessions.

Question and response refinement. Questions were reviewed and revised as follows:

(1) Project staff reviewed the existing questions to make sure that they were worded appropriately. Our criteria for appropriate wording were:
    a. To the extent possible, questions should use the same wording as the sentence in the text that the question was about.
       Example Text: Life must be hard for the people who lived there.
       Example Question: Why must life be hard for the people who lived there?
    b. Questions should not include extraneous words that might confuse students or throw them off track.
       Example Text: The traveler pulled his serape, a colorful woolen shawl, over his shoulders.
       Example Question: Why did he pull his serape over his shoulders?
(2) If an important sentence could not be framed as a why question, the same criteria were applied to the creation of how questions.
    Example Text: I can find out whether there is anything I can do to help my people.
    Example Question: How can he find out whether there is anything he can do to help his people?
(3) Questions, along with follow-up questions (which could be used to clarify questions with ambiguous referents or that were overly long), and expected responses were entered into a grid for each text.
(4) As needed, the questions and responses, along with the texts, were sent to co-PI van den Broek, who provided comments using track changes, including edits to existing questions or suggestions for different questions.

Once final questions were selected, they were inserted into the Question and Response Template (see example in Appendix B).

Results

Table 7 shows the final texts selected for the intervention, and the readability and Text Easability Assessor dimensions of interest. Each text included 4-5 questions.

Table 7. ***Text Dimensions of Study Passages***

Used For	Type	Text-Title	Word Count	Flesch-Kincaid	Narrativity	Syntactic Simplicity	Word Concreteness	Referential Coherence	Deep Coherence
Fall Trial	Narrative	Little Singing Frog Part 1	338	3.7	85%	51%	99%	52%	55%
Fall Trial	Narrative	Little Singing Frog Part 2	334	3.4	85%	53%	97%	35%	31%
Fall Trial	Narrative	Little Singing Frog Part 3	362	4	73%	49%	97%	33%	60%
Spring Pilot	Narrative	A Different Aladdin Part 1	444	3.7	94%	69%	18%	64%	68%
Spring Pilot	Narrative	A Different Aladdin Part 2	456	3.1	91%	78%	44%	30%	59%
Spring Pilot	Narrative	A Different Aladdin Part 3	450	4.9	87%	45%	88%	22%	74%
Spring Pilot	Narrative	The Emperor and the Peasant Boy Part 1	355	4	43%	81%	90%	20%	34%
Spring Pilot	Narrative	The Emperor and the Peasant Boy Part 2	377	3.2	82%	73%	36%	13%	10%
Spring Pilot	Narrative	The Emperor and the Peasant Boy Part 3	409	3.7	74%	81%	83%	33%	55%
Pre-post	Narrative	Annie Helps a Dolphin	233	3.6	85%	85%	75%	37%	41%
Pre-post	Narrative	The Dragon in the Forest	244	3.3	82%	74%	98%	33%	44%
Pre-post	Narrative	Sam Saves a Squirrel	242	3	81%	82%	92%	34%	37%
Pre-post	Narrative	The Spider on the Mountain	240	3.5	81%	75%	99%	33%	44%
Spring Pilot	Info	Jane Goodall Part 1	379	5.1	45%	84%	93%	56%	41%
Spring Pilot	Info	Jane Goodall Part 2	363	3	94%	87%	31%	62%	28%
Spring Pilot	Info	Jane Goodall Part 3	397	4.9	67%	93%	89%	38%	82%
Spring Pilot	Info	Olympic Dreams: Part 1	376	5.4	81%	89%	68%	34%	79%
Spring Pilot	Info	Olympic Dreams: Part 2	378	6.4	50%	79%	80%	46%	26%
Spring Pilot	Info	Olympic Dreams: Part 3	324	6	66%	93%	77%	18%	62%
Spring Pilot	Info	Cesar Chavez - Part 1	406	4	58%	99%	83%	35%	56%
Spring Pilot	Info	Cesar Chavez - Part 2	366	4.4	61%	99%	86%	41%	97%
Spring Pilot	Info	Cesar Chavez - Part 3	416	4.2	41%	99%	84%	39%	87%
Pre-post	Info	Helen Keller	240	4.2	83%	98%	73%	91%	57%
Pre-post	Info	Clara Barton	242	3.6	72%	97%	98%	62%	98%
Pre-post	Info	Gutenberg	277	4.3	45%	93%	91%	59%	97%
Pre-post	Info	Walt Disney	237	4.6	61%	90%	93%	31%	95%

DEVELOPING THE INTERVENTION

The purpose of developing the intervention was to identify a promising questioning approach to improving struggling readers' comprehension of narrative fiction and nonfiction text. Our primary aims in Year 3 were to determine whether the intervention was feasible to implement in typical school settings and promising to improve student reading outcomes.

Specific research questions included:

(1) What is the fit and feasibility of the intervention for implementation in school settings?
(2) What is the effect of the intervention on student reading outcomes?
(3) Do subgroups of struggling readers (paraphrasers and elaborators) respond differentially to intervention?

Below, we first describe the 'final' version of the intervention to provide context for the description of its development. Then, we describe six 'Rounds' of iterative development and refinement that led to the current version of the intervention.

Description of the Final Version of the Intervention: "In-the-Text Connections"

Intervention Structure

The intervention was designed to be delivered in small groups of three to five students, three times per week for 20 to 30 min per session. The intervention was intended to be delivered by the regular classroom teacher, in or outside of the regular classroom or by an interventionist (e.g., reading specialist, Title I teacher, special education teacher, ELL teacher).

The intervention consists of 21 lessons. The first lesson is an introduction to the purpose and materials. The following nine lessons consisted of three narrative texts divided into three sections, with one section read per session. The next nine sessions were non-fiction biography texts given in the same format. During the first story (lessons 1-4) the students used highlighters to help make their connections and learn to make connections in their own words. The last two biographies lessons (13-18) gradually faded the questioning and connecting supports. The final lessons (19-20) were designed to apply "In-the-Text Connections" to a regular book without any additional materials. Table 8 shows the overall lesson sequence and objectives.

Table 8. ***Lesson Sequence and Objectives***

Day/Lesson	Text	Objectives
Day 0 Introduction	Sample Story: Jimmy and the New Bike	• Learn the purpose of making In-the-Text Connections. • Find highly-connected, important sentences in a text. • Practice answering questions to make In-the-Text Connections. • Learn about Points and Rules.
Days 1-3 Training	Story: A Different Aladdin	• Review the purpose of making In-the-Text Connections. • Answer questions to make In-the Text Connections. • Practice working with a partner. • Learn to use their own words (Day 3).
Days 4-6	Story: The Emperor and the Peasant Boy	• Work with a partner to answer questions using their own words. • Make In-the-Text connections without highlighting. • Learn to answer "How" questions (Day 6, Question C).

Days 7-9	Story: Little Singing Frog	• Work with a partner to answer questions using their own words.
Days 10-12	Biography: Jane Goodall	• Work with a partner to answer questions using their own words.
Days 13-15	Biography: Cesar Chavez	• Work with a partner to answer questions using their own words. • Generate questions using the last sentence read.
Days 16-18	Biographies: Olympic Champions	• Work with a partner to answer questions using their own words. • Generate questions using the last sentence read. • Make In-the-Text Connections with no questions.
Days 19-20	Book: The Paper Bag Princess	• Make connections to important parts of a text (with no questions).

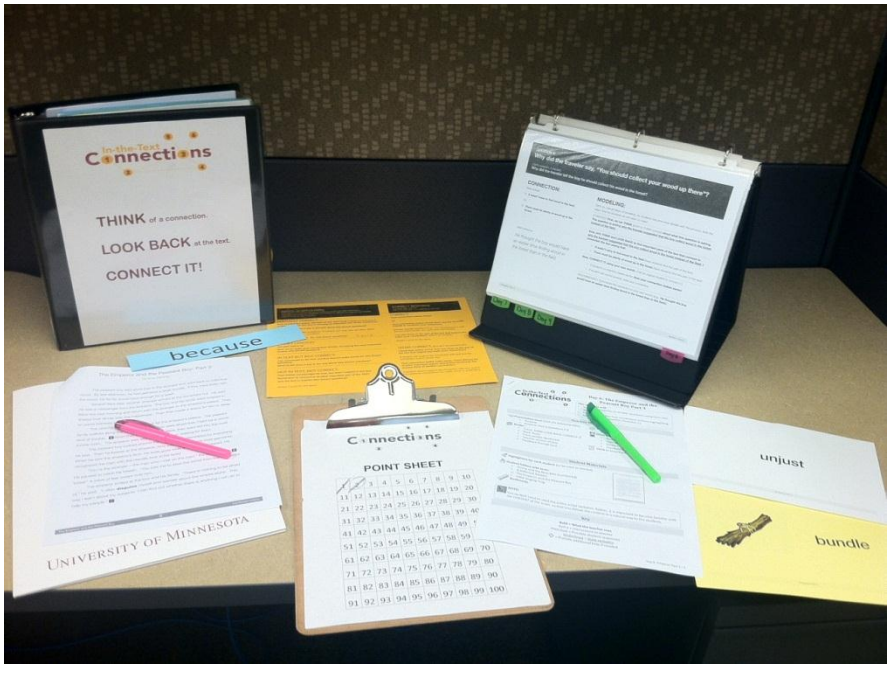
Intervention Materials

Tutoring materials are displayed in Figure 3. Each student had a two-pocket folder with the purpose of the activity written on the front of the folder. Inside the two-pocket folder was a printed text that was divided into a separate part for each intervention session. The text was written in 13 point Helvetica Neue LT Std font on an 8.5 x 11" paper and stapled together into a booklet. At the end of sentences in which a causally-important connection could be made, a highlighted letter was inserted (see sample in Appendix C). The students also had a blue 8 x 1.5" bookmark with the word 'because' on one side to help answer "why" questions and the word "by" to help answer "how" questions on the other side.

The tutor had a three ring binder with the rules displayed on the front of the binder, and the Think-Look back-Connect it! prompt displayed on the back of the binder that contained all the teaching materials. For each lesson the tutor had 8.5 x 5" vocabulary cards, a script for the preteaching component, and teacher modeling as needed. Tutor materials also included a teacher copy of the text and a 'because' bookmark. The teacher copy of the text had underlined words with definitions for those words presented in text boxes in the page margin (see sample in Appendix D). The tutor could provide definitions to the group for the underlined words while reading, as needed.

There was also a presentation easel containing the following materials: rules, the Think-Look back-Connect it! prompt card, question cards that faced the student, and the teacher feedback cards that faced the teacher for each question. The question card had an answer stem to help students answer each question in a complete sentence. To aid the teacher with providing appropriate feedback, tutors were given a feedback prompt card with a modeling script. These feedback prompt and modeling cards were more general than the individual question teacher feedback cards in the binder (see Feedback Prompt Card in Appendix E). There was also a point sheet on a clipboard. The tutor highlighted the goal for each day (to indicate the number of points the group should try to receive) and kept track of points by marking a slash through a number each time a point was earned.

Figure 3. ***Sample Tutoring Materials for Final Intervention Version***



Intervention Protocol

The intervention protocol generally followed the same structure (see "Ongoing" lessons [Days 4 to 12] section). Variation of this structure occurred during the first four training lessons (Days 0 to 3), six lessons that gradually faded the questioning and connecting supports (Day 13 to 18), and two lessons designed to apply "In-the-Text Connections" to a regular book (Day 19 & 20). Below, the intervention protocol for the "on-going script" and the variations (intro, fading, and application) are described.

Four Training Lessons (Day 0 to 3). The four training lessons consisted of Day 0 (introduction) and Day 1 to 3 (question answering with highlighting).

Day 0. The Day 0 script included: (1) introduction of the instructor and students, review of student materials, (2) learning about the purpose, (3) finding important sentences in a story, (4) practicing making in-the-text connections, (5) review of the team points and rules, and (6) a review game. First, for the introduction, each student was asked their name and the instructor introduced him or herself. Then each of the students was given their folder with all the materials. Finally, in the introduction a schedule was provided of the activities for the Day 0 lesson.

Second, to learn about the purpose, tutors facilitated a conversation with the students about the value of *understanding* what they read compared to decoding and fluency. The tutor also described that connecting important parts of the text within the same text ("In-the-Text Connection") can help them understand what they read. These were compared to other connections taught in the schools; text-to-text, text-to-self, text-to-world.

Third, since finding important parts of the story is critical to make an "In-the-Text Connection," the next section explicitly taught students what constitutes an important sentence in a text how to find important sentences in a text. First, the tutor read the story "Jimmy and the New Bike," students followed along on the easel, and the main idea of the story was reviewed (Jimmy wants a bike and works to earn money to get one). Then, the format of the story was

modified from a paragraph format to a numeric list in which each sentence was numbered. For all 11 sentences, a similar type of instructional sequence was followed. The sentence being discussed was in bold-font and arrows were pre-drawn from the sentence being discussed to any other sentences in which this sentence was related to (also termed: "connections"). The students then discussed if the sentence was important (many connections) or not important (few connections). Sentences identified as important were highlighted on the easel by the tutor and by the student on their student copy. After all 11 sentences were discussed the tutor read all highlighted sentences and described how the highlighted sentences provide all the important parts of the story.

Fourth, to practice making "In-the-Text Connections" "Jimmy and the New Bike" was used one more time. "Jimmy and the New Bike" was displayed on the easel in the same format as all the texts in the intervention in regular paragraph format with a bolded boxed letter as stopping points to stop and make an "In-the-text Connection." One student was called onto read the sentence before the black boxed letter A. Then the tutor asked a causal question, modeled the correct answer and emphasized how the answer was an important part of the text (a highlighted sentence). If needed, the tutor repeated the process for letter B.

Fifth, for the team points and rules section, students were taught about the process of earning points and discussed the five rules. It was explained to the students that they earned points for making good "In-the-Text Connections." Then, the rules were discussed and students described examples of each of the rules. Finally, the students and tutor signed the rule page to demonstrate their commitment to the rules.

Last, all students engaged in a review game about the purpose of making "In-the-Text Connections," how to find important parts of the text, how to earn points and the rules. This is how students earned their points for the day and were given their reward sticker.

Day 1 to Day 3. The Day 1 to Day 3 lessons varied slightly from the previously described "on-going" lessons (Day 4 to 12). During these introductory lessons, students were explicitly taught how to: make an "In-the-Text Connection," through teacher modeling, use the "because" bookmark, and share their connection with their partner. Also, to further reinforce the concept of connecting to important parts of the text, students highlighted the important part of the text that supported their connection. At the end of the lesson, all of the highlighted sentences were read aloud and students were prompted to notice how the highlighted portions (sentences identified as important) provided the main idea of the story. Over the course of the three days, the level of support was faded as the students increased their skill and knowledge of how to make an "In-the-Text Connection," use the bookmark, and share their responses with a partner.

"On-going" lessons (Days 4 to 12). A typical intervention session (following training Days 0-3) consisted of two parts: (1) preteaching, and (2) guided practice with feedback. Each intervention session started with goal setting which involved establishing the number of points the group would strive to earn and writing this number on the point sheet.

Preteaching. The preteaching component followed goal setting and was conducted for no more than 5 min. Preteaching involved activating background knowledge and teaching definitions of two to four words that were central to the text. Activating background knowledge consisted of providing students with a brief preview of the text, asking students questions on similar texts they may have read, and/or asking students to discuss related experiences they may have had.

The vocabulary words were highlighted on the student and teacher copies of the text. When introducing the vocabulary words, the tutor first showed the word and/or a picture printed on a card and said the word aloud. Next, the student read the word aloud with the tutor and then pointed to the same word highlighted in the text. Then, the tutor provided a definition for the vocabulary word and/or a picture to depict the meaning of the word. The students repeated the definition back to the tutor. Next, the tutor asked a question pertaining to the word and the student had to respond to the question using the word. This procedure was used to give students several opportunities to actively engage with the vocabulary word. The preteaching component procedure was the same for both feedback conditions.

Guided practice with feedback. The remainder of the session was devoted to reading and making connections. First, the tutor read the text aloud up to the first highlighted letter. The tutor provided meanings to unknown words while reading, as needed. Next, the tutor selected a student to reread the same section. If time was limited, the reading was only done once either by a student or the tutor. Upon reaching the letter, the tutor asked a causal question, which was displayed on the presentation easel, and prompted students to think of a connection, look back to check their connection in the text, and share their connection with a peer. After students their connections with each other, the tutor called on one student to 'Connect it,' and the student shared his/her answer.

The tutor gave structured feedback based on the student's answer. When a student's connection was not correct, tutors attempted to guide them through processing the text. If the connection was still incorrect after two prompts, the tutor provided an explicit model of how to make the connection just as in the direct feedback condition.

The group received points for making connections and for working together and following the rules. The tutor continued the process of reading and answering questions until all questions for the day were answered or till the last 3 minutes of the session. At the end of each session, if the group met their goal, students received stickers to put on their folders.

Gradually fading the questioning and connecting supports (Day 13 to 18). On Days 13-18, the lessons gradually faded the questioning and connecting supports in the following way:

Days 13-15. The questions on the easel were covered up. The tutor explicitly showed the students that, each time they came to a black-boxed letter in the text, the corresponding question was actually just the last sentence they read, turned into a "Why" or a "How" question. The tutor modeled turning a sentence into a "Why" question, and then uncovered the question to check and see if he/she came up with the right question. Students were given the chance to practice with each question, and checked the questions on the easel. They continued to make connections with each question.

Days 16-18. Both the questions and the sentence stem on the easel were covered up. On Day 16, students generated a question using the last sentence they read before the black-boxed letter in the text. On Day 17, the tutor showed them that they could make a connection without even asking a question, by simply re-reading the sentence, saying "because," and then making a connection. The students practiced making connections with no questions on Days 17 and 18.

Applying "In-the-Text Connections" to a regular book (Days 19 & 20). On Days 19 and 20, all of the questioning supports were removed, and the "In-the-Text Connections" approach was applied to a "real" text. Each student received a copy of *The Paper Bag Princess*

by Robert Munsch. The tutor read a few pages of text aloud with the students (using the usual repeated reading format). Then, the tutor modeled identifying an important sentence in the text, and connecting it to an earlier important sentence in the text, using the same procedure as in Days 17-18 (by restating the sentence, saying "because," and making a connection). The tutor was provided with a guide to the important sentences in the text and possible connections (in Appendix F), and students practiced making connections on every page or two, for two sessions (in order to complete the entire book).

Round 11: In-House Development (August-December, 2013)

Below, we describe the iterative development cycle that led to the version of the intervention described above. We begin with "Round 11," picking up from Rounds 6-10 conducted in Year 2.

Project staff reworked the training scripts (Days 0-3) to improve the way the purpose and procedures of In-the-Text Connections were introduced to students. Specifically, we drew from ideas generated in Round 10 (Spring Trial Run of Year 2), in which we added a component to explain that important sentences in a text are connected, and showed students how, when you highlight the most important/highly connected sentences, they basically tell what the text was about. Minor revisions were also made to Days 5-18 to make the scripts more clear and teacher-friendly. Also, we met with a graphic designer, who helped us design a logo and InDesign templates for lesson materials. All materials were revised into a new format to be as user friendly as possible.

Round 12: Fall Trial Run and Reading/Language Arts Observations (November-December, 2013)

In November/December, we conducted a Fall Trial Run of Days 0-3 and observations of participating teachers' reading and language arts instruction.

The purpose of the Fall Trial Run was to determine the feasibility and flow of the revised structure of the intervention. In other words, we examined whether the intervention was operating as intended. We were particularly interested in whether revisions made to Days 0-3 were operating as intended (e.g., the scripts were clear and flowed well; the students understood the highlighting tasks). We also piloted two new measures: a highlighting task and a reading comprehension test (the Multiple-choice Online Cloze Comprehension Assessment; MOCCA). The purpose of the Reading and Language Arts (RLA) Observations was to be able to describe "typical" reading instruction received by all participating students in the study, and to gain a better understanding of how the intervention would fit into the regular reading and language arts instructional programs in schools.

Participants

The Fall Trial Run was conducted with a subset of students who just missed meeting the eligibility criteria for the intervention (n = 7). The "just missed" eligibility criterion was a CBM maze score in the 15.5-16.5 range. Students were placed into two groups of 3-4 students.

The RLA observations were conducted in classrooms of all participating 4th-grade teachers, except for one classroom (from which only 1 student was identified as eligible for the intervention). The classroom teacher indicated she conducted lessons in the same way as another participating teacher; therefore, we skipped this classroom due to scheduling challenges. Thus, 19 classroom teachers were observed.

Procedures

Fall Trial Run. The trial run was conducted for two weeks (6 sessions) in November by two GRAs with experience delivering the intervention in prior years of the project.
The tutors who implemented the Fall Trial Run maintained Intervention Logs of each session that included the total number of minutes of instruction, student attendance, and the number of questions answered. On Day 0, tutors responded to 11 prompts using a 4-point Likert-type scale assessing their agreement (1=disagree to 4=agree) on topics covering perceived student understanding, student motivation, and materials. On the remaining days, tutors responded to 6 to 8 prompts using a 4-point Likert-scale to assess the utility of the highlighting approach, appropriateness of the text and questions, and perceived student motivation. On select days, tutors were asked to note specific issues/problems and suggestions for improvement for items rated lower than 4. Tutors brought their intervention logs to lab meetings to facilitate discussions regarding whether the intervention was operating as intended or additional revisions were needed. Mean ratings were obtained for each component, and open-ended feedback was summarized and used for further revision to the intervention. The Fall Trial Run Intervention Log is found in Appendix G.

During the trial run, members of the research team who were not serving as tutors (PI, project coordinators, 3 GRAs) observed each tutor at least once and took detailed field notes to provide feedback during project staff meetings. At the end of each week, all research staff met to discuss the intervention, using the Intervention Logs and field notes to guide discussion. Detailed notes were taken during these meetings, and Intervention Log data, including ratings and comments, were summarized. Each category of feedback was reviewed and incorporated into revisions of the intervention scripts and materials. Revisions were reviewed by co-PI Espin. After Days 0-3 were piloted, additional research staff assisted in piloting some of the posttest measures. Specifically, we piloted the "near transfer" tasks (described under measures) to see how long they would take with the new Highlighting task, to make sure the Highlighting task was clear to students, to make sure the MOCCA directions were clear, and to determine how students would respond to this test.

RLA observation. The PI and two GRAs with classroom teaching experience and expertise in reading conducted the RLA observations. We asked each teacher to identify a day and time in which we could observe their entire RLA block, and asked specifically to see a "typical" instructional block (in which the structure and activities were similar to what they would do on a typical day). We used a Literacy Checklist (Appendix H) to guide our observations, such that we could identify specific areas of focus (e.g., instruction in fluency, instruction in vocabulary, development/activation of background knowledge, instruction in comprehension strategies, time for independent reading practice) and other elements of instruction (e.g., student engagement, types and levels of texts used, technology use). We noted the amount of time spent on each activity observed and took detailed field notes. These field notes were then typed up into a one-page summary of what was observed and emailed to the teacher to verify whether the observation was accurate. All teachers replied that the observations were accurate representations of their typical instruction.

Results

Fall Trial Run. Based on the tutor's implementation and other research staff members' observations of Days 0-3, revisions were made to scripts and materials. Specifically, scripts were revised to further reduce teacher talk, to clarify which materials to use at what time, and to clarify when and how to provide additional help to students. Materials were revised to reduce the burden on the teacher regarding switching between the scripts and information on the back of the easel. After initial revisions were made, multiple research staff read the scripts aloud in pairs to make sure the revisions achieved the intended purposes (to simplify and reduce the burden on the teacher). Based on research staff members' administration of the Highlighting and MOCCA measures, administration instructions were further revised to clarify the task for students.

RLA Observations. The 19 classrooms were observed for an average of 88 min (range = 50-120 min). Most RLA blocks included an introduction of the lesson focus and objectives/learning targets, followed by a combination of whole class instruction (observed in 18 classrooms), small-group instruction with push-in or pull-out from a specialist (observed in 8 classrooms), small-group instruction with no push-in/pull-out (observed in 7 classrooms), independent reading and/or writing activities (all classrooms), peer-mediated activities (6 classrooms), and technology use (6 classrooms; typically this involved independent reading games on the computer). In four classrooms, teachers met with individual students for fluency checks (2), guided reading activities (1), or a writing conference (1). A range of fiction and nonfiction texts were used, with 5 classrooms using only fiction, 3 classrooms using only nonfiction, and 11 classrooms using both.

Teachers addressed reading and language arts skills in four general areas: vocabulary and language (observed in most classrooms), reading comprehension (observed in most classrooms), guided reading (observed in most classrooms), writing (observed in some classrooms), and word study (observed in few classrooms). Specific skills or strategies addressed within each category are displayed in Table 9.

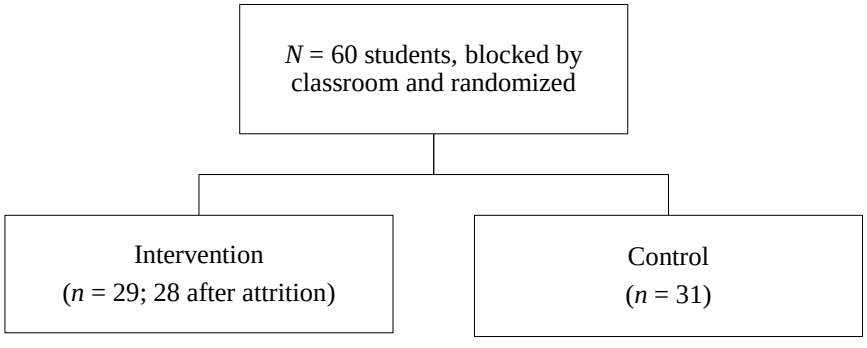
Table 9. ***Focus of Learning Activities during Reading and Language Arts Observations***

Learning Focus	Types of Skills/Strategies Addressed
Vocabulary and language	• defining words • identifying parts of speech • synonyms & antonyms • using picture cues • personification and figurative language (idioms, similes, metaphors) • identifying interesting words in text
Reading Comprehension	• building schema • identifying elements of character descriptions • identifying story elements • identifying informational text structures • identifying text features • making predictions • making inferences • comparing/contrasting • question-generating and question-answering before, during, after reading • making connections (text to text, text to self, text to world) • visualizing • recognizing author's purpose • identifying how point of view affects how a story is told • identifying main ideas and details to summarize a text • identifying main points of argumentative writing • using graphic organizers • close reading of text (look for unknown words, key points, connections and questions)
Guided Reading	• small group reading: including previewing vocabulary, reading aloud in round robin fashion or silent reading, and questioning at varying levels of explicitness
Writing	• writing letters • writing dialog • descriptive writing • character webs • writing personal narratives • science fiction writing • including details in procedural writing • poetry writing • writer's workshop
Word Study	• homophones • pattern word sorting • prefixes and suffixes

Round 13: Spring Pilot (January through March 2014)

The purpose of the Spring Pilot was to obtain preliminary data regarding the extent to which the revised Causal questioning intervention showed promise for subgroups of struggling comprehenders, and whether the intervention was feasible to implement by school-based personnel. We used an experimental, randomized control trial, as illustrated in Figure 4.

Figure 4. ***Assignment to Feedback Condition and Text Type***



Participants and Group Assignment

Students. All students identified as eligible for intervention (n = 60) were blocked within classroom and assigned randomly to intervention (n = 29) or control (n = 30). One student assigned to intervention was withdrawn following random assignment, when the school determined that she was more in need of a fluency intervention. Intervention students were placed in groups of 3-5 students within their schools based on availability of students and teachers. Demographic and screening data for students in the two groups are provided in Table 10, along with chi-square and F-statistics. There were no statistically significant differences between the two groups in terms of demographic variables.

Table 10. ***Student Demographics***

	Total (n = 59)		Intervention (n = 28)		Control (n = 31)			
	n	%	*n*	%	*n*	%	X^2	*p*
Sex (M)	37	63%	16	57%	21	65%	.707	.401
ELL	22	37%	11	39%	11	35%	.091	.763
Free/Reduced Lunch	40	68%	17	61%	23	74%	1.223	.269
Race							4.203	.649
American Indian	1	2%	0	0	1	3%		
African American	20	34%	8	29%	12	39%		
Asian	5	8%	2	7%	3	10%		
Hispanic	17	29%	9	32%	8	26%		
White	14	24%	8	29%	6	19%		
Other	3	5%	1	4%	2	6%		
Special Education	6	10%	4	14%	2	6%	.988	.320

	Total (n = 59)		Intervention (n = 28)		Control (n = 31)		X^2	p
	Mean	SD	Mean	SD	Mean	SD	F(1, 60)	P
Age (end of year)	10.19	.27	10.19	.28	10.19	.28	.004	.953

Teachers/tutors. Teachers and specialists from participating schools were invited to implement the intervention. Project staff met with staff at each school to discuss who would be the implementers. In four schools (two in District 1 and two in District 2), school staff agreed to implement, but the specific type of staff member varied by school. In both schools in District 1, a classroom teacher agreed to implement. In one school in District 2, an intervention specialist agreed to implement; in the other school, two classroom teachers decided to split implementation (such that one implemented for the first half of the intervention period, and the other implemented for the last half).

In the remaining three schools, project staff implemented the intervention, because the schools did not feel they had the resources to allow school staff to implement the small group intervention for 30 min, 3 times per week. In these schools, project staff members who had been classroom teachers in the past were assigned to deliver the intervention. Table 11 provides a summary of the demographics of teachers and tutors who delivered the intervention.

Table 11. ***Teacher and Tutor Demographics***

	Teachers (school staff; n = 5)		Tutors (project staff; n = 4)	
	n	%	n	%
Sex (female)	4	80	3	75
Race				
White	4	80	4	100
Hispanic	1	20	0	0
Age				
20-29	0	0	2	50
30-39	0	0	2	50
40-49	3	60	0	0
50-59	1	20	0	0
60 +	1	20	0	0
Highest Degree				
Bachelor's	1	20	1	25
Master's	1	20	0	0
Master's + Additional Coursework	3	60	3	75
	M (range)	SD	M (range)	SD
Years in current position	6.60 (2-12)	3.84	2.25 (1-3)	0.96
Years in general education	22.20 (14-36)	8.96	1.75 (0-6)	2.87
Years teaching Grade 4	9.20 (0-22)	8.44	0	0
Years teaching special ed.	2.60 (0-13)	5.81	2.50 (0-5)	2.38
Hours of PD in reading comprehension	15.80 (4-32)	12.74	10.00 (0-30)	17.32

Measures

Several types of measures were used during the Spring Pilot, including measures of fit and feasibility, fidelity, and promise of the intervention.

Fit and feasibility. Fit and feasibility feedback was solicited from teachers/tutors who implemented the intervention and from intervention specialists. Teachers maintained logs of each session that included the total number of minutes of instruction, student attendance, and the number of questions answered. These data were compiled for both group and individual dosage information (total sessions completed, total minutes completed, and total questions answered). Space was also provided for tutors to note any special issues related to individual student attention and behavior, as well as any special issues related to the session such as what went well, didn't go well, suggestions for improvement, and corrections needed for the scripts and materials. A Spring Pilot Intervention Log is in Appendix I.

Intervention fidelity. The fidelity observation form was revised based on findings from Year 2. Specifically, we included the "Scaffolded" feedback approach that appeared to be most feasible/promising from Year 2 results. Also, we added items to better capture the extent to which students were engaging in the activities and responding to teacher prompts and feedback, and we added a "Wrap-Up" section that previously had not been included on the form. The final version of the Fidelity Form (in Appendix J) includes the sections listed below. Fidelity was calculated as the percentage of items observed in each section, a "general fidelity" percentage (including set-up, introduction, delivery, and wrap-up), and "In-the-Text Connections" percentage (including preteaching, reading and connecting, and feedback), and an "overall fidelity" percentage (including all items).

- Set-up (on time, materials ready; 3 items)
- Introduction (purpose, review how to earn points, set goal; 3 items)
- Preteaching (introduce text, preview/review text, introduce key vocab; 5 items)
- Delivery (familiarity with and fluent delivery of script, maintaining student engagement, reinforcement of desired student behaviors; 4 items)
- Reading and connecting (teacher reads, students read, students Think-Look Back-Connect It, share connections with partners and group; 9 items)
- Feedback (teacher provides praise, scaffolding and modeling as needed; 8 items)
- Wrap-up (read to end, predict/state opinion, check goal; 3 items)

Control "fidelity." The fidelity form described above was used to observe control students who were receiving other interventions in their schools. The purpose of these observations was to determine the extent to which there was overlap between the intervention and control instruction.

Teacher/tutor feedback. Teachers and tutors completed a questionnaire after the Spring Pilot was completed. This questionnaire covered a wide range of topics covering all aspects of the intervention. There a total of 50 items in the survey, addressing questions related to teachers' and tutors' perceptions of the intervention and its components.

A total of 10 sections of questions were asked as following: (1) intervention preparation, (2) perceptions of intervention as a whole, (3) perceptions of introductory lessons, (4) perceptions of the reading format throughout the intervention, (5) perceptions of pre-teaching,

(6) perceptions of questions and feedback, (7) perceptions of motivation/engagement strategies, (8) perceptions of intervention structure, (9) perception of student responsiveness, and (10) perceptions of intervention components. The survey primarily contained close-ended questions with only five open-ended question requesting additional comments at the end of two sections, "Intervention Preparation" and "Perceptions of Intervention Components." These questions asked the tutor to reflect on their perceptions of various aspects of the intervention. For intervention preparation, the survey measured the reported implementation of preparation time, and all of the data were descriptive.

In addition to the perceptions of the intervention structure and the perception of student responsiveness, which focused on the perception level of agreement, the rest of the questions mainly measured the perceived level of helpfulness. For example, interventionists were asked to indicate how helpful for students each of the following was (i.e., encouraging students to use their own words to make connections). The interventionists' replies were on a three-point Likert scale, in which 1=unhelpful, 2=somewhat helpful, and 3= very helpful. Perceptions of intervention components were the only two questions that asked interventionists to rank their preferences for the intervention components as well as asked teachers and tutors to discuss any changes they would make, and how these changes could be implemented. The tutor questionnaire is in Appendix K.

Promise. Measures of the "promise" of the intervention to improve student reading outcomes were given at pre-, mid-, and posttest, as shown in Table 12. "Near transfer" tasks refer to tasks administered with texts under reading and questioning conditions very similar to the intervention. "Far transfer" tasks refer to tasks administered with texts with no questioning.

Table 12. ***Pre-, Mid-, and Posttest Measures***

Test Phase	Measure	Text Used
Pretest	Near Transfer (Story) • questions with prompting • recall • oral reading • highlighting	Near Transfer (Story): The Dragon in the Forest/The Spider on the Mountain
	Far Transfer (Story and Biography) • recall • oral reading	• Far Transfer (Story): Annie and the Dolphin/Sam Saves a Squirrel • Far Transfer (Biography): Walt Disney/Johannes Gutenberg
Mid-test	Near Transfer (Story and Biography) • questions with prompting • recall • oral reading • highlighting	• Near Transfer (Story): The Dragon in the Forest/The Spider on the Mountain • Near Transfer (Biography): Helen Keller/Clara Barton
Posttest	Near Transfer (Biography) • questions with prompting • recall • oral reading • highlighting	Near Transfer (Biography): Helen Keller/Clara Barton
	Far Transfer (Story and Biography) • recall • oral reading	• Far Transfer (Story): Annie and the Dolphin/Sam Saves a Squirrel • Far Transfer (Biography): Walt Disney/Johannes Gutenberg
Posttest Only	Maze MOCCA	

Pre-, mid-, and posttest texts. Near transfer texts were administered in a similar format to the format used during training (with the same types and timings of questions asked), and far transfer texts were administered without the questioning. Including near- and far-transfer texts allowed us to examine immediate effects of the questioning intervention, as well as whether intervention effects transferred when the intervention was no longer provided. Near- and far-transfer texts included both stories and biographies, to reflect the types of texts used during intervention. The stories were designed to be parallel to each other, in that they followed a very similar story structure and general plot with similar goals, sub-goals, and resolutions, but with different characters and specific scenarios. The informational texts were similar in length, readability, and biographical structure. All texts were similar to each other and to texts used in the intervention in terms of readability and the five Text Easability features.

Near transfer stories ("The Dragon in the Forest" and "The Spider on the Mountain") were counterbalanced and administered at pre- and mid-test (immediately before and after students received the intervention using stories). Near transfer biographies ("Helen Keller" and "Clara Barton") were counterbalanced and administered at mid- and posttest (immediately before and after students received the intervention using biographies).

Far transfer stories ("Annie and the Dolphin" and "Sam Saves a Squirrel") were counterbalanced and administered at pre- and posttest. Far transfer biographies ("Walt Disney" and "Johannes Gutenberg") were counterbalanced and administered at pre- and posttest. Administration instructions are in Appendices L-N. Examiner versions of the pre-mid-posttest protocols are in Appendices O-S.

Dependent variables. The following dependent variables were measured using near and far transfer texts:

Responses to questions using intervention approach (near transfer only). The student read the text aloud, responding to four causal online questions with prompting and feedback similar to that delivered during the intervention. Scores included (a) number of questions answered correctly before prompting, (b) number of questions answered correctly after prompting, and (c) number of prompts needed (maximum = 2 per question).

Verbal recalls (near and far transfer). After reading and responding to questions (for the near transfer text) and after reading only (for the far transfer text), the student was asked to recall the text just read. Specifically, the students were prompted, "Now, let's pretend I am a friend from your class who didn't get to read this story. Please tell me the story about [read title] in your own words. Go ahead and start." The student was additionally prompted, "What happened next?" as needed until the student finished retelling. If the student did not tell the very end of the text, the student was finally prompted, "Is there anything else about the text that you'd like to tell me?"

Students' responses during the recall of each text were audio recorded, transcribed, and parsed into thought units based on the original text. Each recalled thought unit was compared to the parsed story units in the original text and coded using a scheme adapted from Kendeou and van den Broek (2005). The coding included the original story unit number to which it matched (if any), and the extent to which it matched or was constrained by the text. The following five codes were recorded:

- Conservative (C; directly matches a story unit and is a literal or nearly literal rendering of the original story unit)

- Liberal (L; directly matches a story unit and captures the gist of the story unit (the essence of its meaning; paraphrasing may be present)
- Matched but incorrect (MI; directly matches a story unit, but is significantly incorrect. MI is reserved for absolutely and importantly contradictory interpretations, not just an erroneous detail of the story)
- No match consistent (NMC; cannot be matched directly with the gist meaning of any one or two adjacent story units, but is valid, consistent with/constrained by the information in the text, or plausible.)
- No match inconsistent (NMI; invalid, inconsistent (unconstrained by) the information in the text. The thought unit should be flat-out incorrect to be NMI)

The number of matches to highly connected original story units was also counted. Highly connected story units were those that had at least six or seven (depending on the text) connections to other story units, or stated a goal or attainment of a goal in the story. The recalls were also coded for order of recall, which was the correlation of the sequence of recalled thought units with sequence of the original text story units.

Words read correctly in one min (near and far transfer). After completing the verbal recall of the near or far transfer text, students re-read the text aloud for one min, following the same procedures used during the CBM oral reading task described above. WRC were recorded.

Proportions of highly-connected sentences identified through highlighting (near transfer only). After completing the oral reading task on the near transfer texts only, the student was asked to "highlight NO MORE THAN 8 sentences that you think include the MOST IMPORTANT parts of the story – the parts you really need to remember to understand what the story was about." The purpose of this task was to determine the extent to which students identified the highly-connected parts of the text as "important." A limit of 8 sentences was provided to ensure that students had ample opportunity to capture the highly-connected parts of the text (which spanned about 7 sentences in each text), without given them freedom to highlight the entire text. If students highlighted more than 8 sentences, they were asked to cross out less-important sentences. Two scores were recorded: (a) the proportion of highly-connected story units that the student identified as important, and (b) the proportion of sentences that the student highlighted that were highly connected.

Procedures

Teacher training and support. A day-long workshop was provided to all teachers and tutors who delivered the intervention. The workshop consisted of (a) describing the objectives, purpose, and theoretical foundation of the study and the intervention; (b) providing an overview (including video) of the intervention and materials; (c) demonstration of how to teach students to make In-the-Text Connections; (d) demonstration and practice in how to deliver the intervention; (e) demonstration and practice in how to give feedback; and (f) time for planning the intervention schedule.

During the workshop, teachers sat in groups such that there was at least one teacher, one tutor with experience delivering the intervention, and one other research team member who was assigned to be an "intervention partner" with the participating teachers (described below). Many opportunities to practice, ask questions, and discuss the intervention procedures in these small groups were provided throughout the workshop. This way, the experienced tutors could model

procedures, share tips for implementation, and answer questions in a setting that was designed to be non-threatening and comfortable for the teachers.

Each participating teacher was assigned an Intervention Partner to support the teacher during implementation. The role of the Intervention Partner was described to the teachers as follows: (a) help you practice and prepare to deliver the intervention, (b) attend your intervention sessions at least once per week, (c) answer questions and help with problem solving, (d) provide any needed materials, (e) provide modeling and feedback as needed, and (f) serve as a liaison between you and the UMN team. The teachers and Intervention Partners were given time at the end of the workshop to decide how they would work together, and to set a schedule.

Following training, the research team met weekly for 1-2 hours. Part of these meetings was devoted to problem solving: tutors and Intervention Partners raised issues they encountered or observed during intervention sessions, such as behavior problems, maintaining student engagement, helping students understand the prompts, etc. The group discussed and generated ideas for addressing these issues. Intervention Partners were to communicate these ideas back to their teachers.

Intervention implementation and testing. Implementation began in late January and lasted through the end of March. All intervention and control students were pretested individually by research staff who were NOT delivering the intervention to them and who were blind to their intervention condition. Then, each teacher/tutor began with Day 0, followed by Days 1-9 using the intervention stories. Intervention was delivered for approximately 30 min per session, 3 times per week. After the first 3 weeks of intervention, all intervention and control students were individually administered the mid-tests (again, by staff not delivering the intervention to them and who were blind to their intervention condition), which took approximately one week (no intervention was implemented at this time). When mid-testing was complete, the teacher/tutor implemented Days 10-18 using intervention biographies. Then, intervention and control students were individually administered the posttests (again, by staff not delivering the intervention to them, and blind to their intervention condition).

Fidelity observations. During the Spring Pilot, two researchers observed 2 out of 8 intervention groups (25%) together to obtain interrater agreement. Interrater agreement, calculated as the number of agreements divided by the number of agreements plus disagreements, was 92.36%. After establishing interrater agreement, the two researchers observed each tutor between Days 5-9 (with most on Days 6-7). Immediately following these observations, feedback was given to the teachers/tutors with the purpose of improving implementation fidelity. Fidelity observations were conducted again between Days 12-15.

Control fidelity observations. Control observations were conducted by the PI toward the end of the intervention period. The PI requested to observe any intervention received by control students that included a focus on reading comprehension. Interventions included small-group guided reading instruction led by the classroom teacher or an ELL or reading intervention specialist, and typically lasted between 20-30 minutes. The PI used the fidelity form to try to capture any overlap with the In-the-Text Connections intervention, and took detailed field notes as well.

Recall coding training and reliability. Recall coding guidelines from previous studies were revised for the current study. During training and practice, the team revised and refined glossaries of exemplary coded thought units for each text.

The glossaries and the procedure document were the basis of GRA training on recall coding. Conducting training of GRA coders and practicing by the whole team took approximately two weeks. The team discussed and practiced applying the coding procedure as a group to five to eight protocols per story. Then, partners of GRAs and project coordinators practiced coding five protocols jointly and the PI performed IRA for each coding pair. . Questions were discussed and resolved at weekly team meetings by the PI or by the coding experts; examples were added to glossaries. When pairs achieved at least 75% IRA on codes and on story unit numbers, the GRAs in the pair were approved to code independently. After coding reliability was demonstrated, pairs coded one protocol in common for every two protocols coded independently. The PI made IRA checks on independent and consensus-coded protocols. Ten percent of protocols for each individual were selected randomly for IRA. Selection of 10% of each set of protocols assigned to individuals or pairs was repeated for IRA until all coding was complete.

The IRAs on recall coding were calculated for agreement on codes and agreement on story unit numbers of conservative or liberal matches. Note that until the third party IRA with an individual or a pair was at least 75%, PI coding and IRA checking continued. Coding is still in process, so final IRA percentages are not yet available.

During the two and a half months of coding, the PI, PCs, and GRAs met weekly to practice coding and/or review any questions on coding to maintain calibration.

Data coding and entry. All data were transcribed (as needed), coded, and entered into an excel spreadsheet. All relevant codes were counted, and counts were double-checked by a second coder. All data were entered by pairs of GRAs, who entered all data twice into two separate excel worksheets. For the first entry, one GRA called out the data and the other entered the data. For the second entry, the GRAs switched roles. The two worksheets were then compared using the logical test formula in Microsoft excel, and all discrepancies were resolved. Data were imported to SPSS 21.0 for analysis.

Results

Research Question 1: What is the fit and feasibility of the intervention for implementation in school settings?

Fidelity. Fidelity observations were conducted to determine whether the tutors implemented the intervention as intended. Whereas this information is important for interpreting student response to the intervention, it is also important to know whether the teachers/tutors were able to implement the intervention based on the training and support that they received to implement it. Table 13 shows the mean proportion of fidelity with which the intervention was implemented at Time 1 and Time 2 within each fidelity category. Generally, teachers/tutors implemented all versions of the intervention with high levels of fidelity, suggesting it was feasible to implement accurately.

Table 13. ***Fidelity Percentages at Time 1 and Time 2***

	Teachers (*n*= 4)		Tutors (*n*= 4)		Total	
Measures	Mean (%)	SD	Mean	SD	Mean	SD
Set-up (on time, materials ready; 3 items)						
Time 1	91.67	16.67	100.00	0.00	95.83	11.79
Time 2	91.67	16.67	100.00	0.00	95.83	11.79
Introduction (purpose, review how to earn points, set goal; 3 items)						
Time 1	66.67	27.22	83.33	19.25	75.00	23.57
Time 2	83.33	19.25	91.67	16.67	87.50	17.25
Preteaching (introduce text, preview/review text, key vocab, students respond; 5 items)						
Time 1	88.89	15.71	88.89	15.71	88.89	14.55
Time 2	94.44	11.11	94.44	11.11	94.44	10.29
Delivery (familiarity and delivery of script, engagement, reinforcement; 4 items)						
Time 1	91.67	16.67	93.75	7.98	92.71	12.15
Time 2	91.67	11.79	93.75	7.98	92.71	9.38
Reading and connecting (teacher reads, students reads, share connections with partners; 9 items)						
Time 1	95.97	4.87	88.10	16.19	92.03	11.84
Time 2	92.33	9.50	99.44	1.11	95.89	7.33
Feedback (teacher provides praise, scaffolding and modeling as needed; 8 items)						
Time 1	96.67	6.67	94.89	6.34	95.78	6.10
Time 2	89.58	12.50	100.00	0.00	94.79	9.90
Wrap-up (read to end, predict/state opinion, check goal; 3 items)						
Time 1	100.00	0.00	100.00	0.00	100.00	0.00
Time 2	83.33	33.33	100.00	0.00	91.67	23.57
Fidelity General (set-up, intro, delivery, wrap-up)						
Time 1	89.17	12.51	94.05	2.38	91.61	8.74
Time 2	89.23	15.21	95.24	5.50	92.23	11.07
Fidelity ITT (preteaching, reading & connecting, feedback)						
Time 1	95.17	2.13	90.04	9.90	92.60	7.17
Time 2	92.14	3.47	99.06	1.19	95.60	4.41
Overall Fidelity						
Time 1	93.64	3.53	91.00	7.37	92.32	5.53
Time 2	91.05	4.61	98.18	1.29	94.61	4.93

Control fidelity. Whereas control intervention components overlapped somewhat with the *set-up, introduction, pre-teaching,* and *delivery* components of the In-the-Text Connections intervention, there was very little overlap with the *reading and connecting* and *feedback* components of the intervention. Teachers did often read aloud to students and/or use round-robin read-aloud approaches, and inserted questions and discussion into these read-alouds; however, the types of questions were rarely related to making text-based connections, and feedback was minimal. A more extensive analysis of the overlap between the In-the-Text Connections intervention and control intervention will be conducted during our no-cost extension year.

Teacher/tutor ratings. Overall, teachers and tutors rated the intervention as feasible to implement, with some differences between the two groups. The results of this survey will be more completely summarized in a separate technical report (Hsu, in preparation). Tables 14-25 below summarize the teachers' and tutors' ratings and suggestions for further revision.

Table 14. ***Preparation Time Day 0-18***

Preparation time	*n*	*M (SD)*	*n* (%)	Group	
				Tutor *n* (%)	Teacher *n* (%)
Days 0-3	8	2.38 (0.52)			
15 min or less					
16-30 min			5 (62.5)	4 (100)	1 (25)
31-60 min			3 (37.5)		3 (75)
Days 4-9	8	1.75 (0.46)			
15 min or less			2 (25)	2 (50)	
16-30 min			6 (75)	2 (50)	4 (100)
31-60 min					
Days 10-18	8	1.38 (0.52)			
15 min or less			5 (62.5)	2 (50)	3 (75)
16-30 min			3 (37.5)	2 (50)	1 (25)
31-60 min					

Table 15. ***Perception of intervention as a whole***

Perception	*n*	*M (SD)*	*n* (%)	Group	
				Tutor *n* (%)	Teacher *n* (%)
Help understand the stories	8	3.5 (0.54)			
Somewhat			4 (50)	4 (100)	
Very much			4 (50)		4 (100)
Help understand the biographies	8	3.5 (0.54)			
Somewhat			4 (50)	1 (25)	3 (75)
Very much			4 (50)	3 (75)	1 (25)
Work best	7	2.43 (0.98)			
Stories			2 (28.6)	1 (33.3)	1 (25)
Biographies					
No difference			5 (71.4)	2 (66.6)	3 (75)

Table 16. ***Perception about the introductory lessons***

Perception	*n*	*M (SD)*	*n* (%)	Group	
				Tutor *n* (%)	Teacher *n* (%)
Highlighting	8	2.88 (0.35)			
Unhelpful					
Somewhat helpful			1 (12.5)	1 (25)	
Very helpful			7 (87.5)	3 (75)	4 (100)
Modeling	8	2.88 (0.35)			
Unhelpful					
Somewhat helpful			1 (12.5)	1 (25)	
Very helpful			7 (87.5)	3 (75)	4 (100)
Using own words	8	2.63 (0.74)			
Unhelpful			1 (12.5)	1 (25)	
Somewhat helpful			1 (12.5)	1 (25)	
Very helpful			6 (75)	2 (50)	4 (100)

Table 17. ***Perception about pre-teaching components***

Perception	*n*	*M (SD)*	*n* (%)	Group	
				Tutor *n* (%)	Teacher *n* (%)
Reminding about the purpose	9	2.11 (0.33)			
Unhelpful					
Somewhat helpful			8 (88.9)	4 (100)	4 (80)
Very helpful			1 (11.1)		1 (20)
Previewing the text	9	3 (0)			
Unhelpful					
Somewhat helpful					
Very helpful			9 (100)	4 (100)	5 (100)
Teaching key vocabulary	9	2.78 (0.44)			
Unhelpful					
Somewhat helpful			2 (22.2)	2 (50)	
Very helpful			7 (77.8)	2 (50)	5 (100)

Table 18. ***Perception about specific instructional features***

Perception	*n*	*M (SD)*	*n* (%)	Group	
				Tutor *n* (%)	Teacher *n* (%)
Repeated Reading	9	2.78 (0.44)			
Unhelpful					
Somewhat helpful			2 (22.2)	1 (25)	1 (20)
Very helpful			7 (77.8)	3 (75)	4 (80)
Using the prompt	9	2.56 (0.53)			
Unhelpful					
Somewhat helpful			4 (44.4)	3 (75)	1 (20)
Very helpful			5 (55.6)	1 (25)	4 (80)
Making In-the-Text Connections	9	3 (0)			
Unhelpful					
Somewhat helpful					
Very helpful			9 (100)	4 (100)	5 (100)
Partner sharing	9	2.44 (0.73)			
Unhelpful			1 (11.1)	1 (25)	
Somewhat helpful			3 (33.3)	2 (50)	1 (20)
Very helpful			5 (55.6)	1 (25)	4 (80)
Providing modeling	9	2.89 (0.33)			
Unhelpful					
Somewhat helpful			1 (11.1)	1 (25)	
Very helpful			8 (88.9)	3 (75)	5 (100)
Generating students' own questions	8	2.75 (0.43)			
Unhelpful					
Somewhat helpful			2 (25)	2 (50)	
Very helpful			6 (75)	2 (50)	4 (100)
Making connections without asking questions	8	2.63 (0.52)			
Unhelpful					
Somewhat helpful			3 (37.5)	2 (50)	1 (25)
Very helpful			5 (625)	2 (50)	3 (75)

Table 19. ***Perception about scaffolding feedback***

Perception	*n*	*M (SD)*	*n* (%)	Group	
				Tutor *n* (%)	Teacher *n* (%)
Correct responses	9	2.56 (0.53)			
Unhelpful					
Somewhat helpful			4 (44.4)	3 (75)	1 (20)
Very helpful			5 (55.6)	1 (25)	4 (80)
Partially correct responses	9	3 (0)			
Unhelpful					
Somewhat helpful					
Very helpful			9 (100)	4 (100)	5 (100)
Overly lengthy answers	8	2.87 (0.35)			
Unhelpful					
Somewhat helpful			1 (12.5)	1 (25)	
Very helpful			7 (87.5)	3 (75)	4 (100)
In the text but not correct answers	8	3 (0)			
Unhelpful					
Somewhat helpful					
Very helpful			8 (100)	4 (100)	4 (100)
Not in the text and not correct answers	9	2.89 (0.33)			
Unhelpful					
Somewhat helpful			1 (11.1)		1 (20)
Very helpful			8 (88.9)	4 (100)	4 (80)

Table 20. ***Perception about motivation/engagement strategies***

Perception	*n*	*M (SD)*	*n* (%)	Group	
				Tutor *n* (%)	Teacher *n* (%)
Setting a goal	8	2.13 (0.84)			
Unhelpful			2 (25)	2 (50)	
Somewhat helpful			3 (37.5)	1 (25)	2 (50)
Very helpful			3 (37.5)	1 (25)	2 (50)
Giving points	8	2.25 (0.89)			
Unhelpful			2 (25)	2 (50)	
Somewhat helpful			2 (25)	1 (25)	1 (25)
Very helpful			4 (50)	1 (25)	3 (75)
Giving stickers	8	2.63 (0.74)			
Unhelpful			1 (12.5)	1 (25)	
Somewhat helpful			1 (12.5)	1 (25)	
Very helpful			6 (75)	2 (50)	4 (100)
Giving verbal praise	9	3 (0)			
Unhelpful					
Somewhat helpful					
Very helpful			9 (100)	4 (100)	5 (100)

Table 21. ***Frequency of using motivation/engagement strategies***

Items	*n*	*%*	Ranking
Repeat In-the-Text Connection rules	4	44.4	3
Generate new rules	-	-	-
Redirect off-task students	9	100	1
Praise on-task students	6	66.7	2
Other	1	14.3	4

Table 22. ***Perception about intervention structures***

Perception	*n*	*M (SD)*	*n* (%)	Group	
				Tutor *n* (%)	Teacher *n* (%)
5 questions within one session	9	3.67 (0.50)			
Disagree					
Agree			3 (33.3)		3 (60)
Strongly agree			6 (66.7)	4 (100)	2 (40)
Group Size	9	2.89 (0.78)			
Disagree			3 (33.3)		3 (60)
Agree			4 (44.5)	3 (75)	1 (20)
Strongly agree			2 (22.2)	1 (25)	1 (20)
30 minutes a session	9	3.44 (0.88)			
Disagree			2 (22.2)		2 (40)
Agree			1 (11.1)	1 (25)	
Strongly agree			6 (66.7)	3 (75)	3 (60)
3 times a week	9	3.44 (0.53)			
Disagree					
Agree			5 (55.6)	2 (50)	3 (60)
Strongly agree			4 (44.4)	2 (50)	2 (40)
Location Inside class	6	3.67 (0.52)			
Disagree					
Agree			2 (33.3)		2 (40)
Strongly agree			4 (66.7)	1 (100)	3 (60)
Location outside class	9	2.89 (1.27)			
Strongly disagree			2 (22.2)		2 (40)
Disagree			1 (11.1)		1 (20)
Agree			2 (22.2)	1 (25)	1 (20)
Strongly agree			4 (44.5)	3 (75)	1 (20)
Helpfulness of the script	9	3.56 (0.53)			
Disagree					
Agree			4 (44.4)	3 (75)	1 (20)
Strongly agree			5 (55.6)	1 (25)	4 (80)
Helpfulness of the materials	9	3.78 (0.44)			
Disagree					
Agree			2 (22.2)	1 (25)	1 (20)
Strongly agree			7 (77.8)	3 (75)	4 (80)

Table 23. ***Perception about students' responsiveness***

Perception	*n*	*M (SD)*	*n* (%)	Group	
				Tutor *n* (%)	Teacher *n* (%)
Asking causal/explanatory (why/how) questions	9	3.78 (0.44)			
Disagree					
Agree			2 (22.2)		2 (40)
Strongly agree			7 (77.8)	4 (100)	3 (60)
Challenges of questions	9	2.89 (0.61)			
Disagree			2 (22.2)	1 (25)	1 (20)
Agree			6 (66.7)	3 (75)	3 (60)
Strongly agree			1 (11.1)		1 (20)
Engagement	9	3.44 (0.73)			
Disagree			1 (11.1)	1 (25)	
Agree			3 (33.3)	2 (50)	1 (20)
Strongly agree			5 (55.6)	1 (25)	4 (80)

Table 24. ***Difficulties and importance of the intervention***

Items	Difficult			Importance		
	n	*%*	Ranking	*n*	*%*	Ranking
Pre-teaching (preview/review of text)	-	-	-	9	100	1
Pre-teaching (vocabulary)	1	11.1	4	9	100	1
Repeated reading (teacher reads/student reads)	1	11.1	4	4	44.4	5
Asking and clarifying the questions	1	11.1	4	8	88.9	2
Reminding students to look back at the text to help make their connections	2	22.2	3	7	77.8	3
Eliciting students' answers	1	11.1	4	9	100	1
Helping students make two-part connections	5	55.6	1	9	100	1
Encouraging students to share with a partner	3	33.3	2	3	33.3	6
Providing appropriate scaffolding	2	22.2	3	8	88.9	2
Providing appropriate modeling	-	-	-	9	100	1
Fading questions	1	11.1	4	5	55.6	4
Maintaining students' engagement	1	11.1	4	8	88.9	2

Table 25. ***Suggestions for the intervention***

Category	Suggestions
Suggestions for introductory lessons	• Suggest explaining the goals more clearly to the students, i.e., the differences or expectations between the beginning lessons and ending lessons about what they would learn • Suggest providing more examples or explanations for the transition, i.e., the highlighting skills they practiced during the intervention is related to help them find the important sentences • Suggest spending more time on explaining or practicing using students' own words to answer the questions • The transition between Day 3 and Day 4 could balance
Overall suggestions	• Add additional sessions by using real books to generate their own questions and connections • Explain the final goal of the intervention to the students at the first beginning (i.e., be able to make their own connections by generating their own questions) • Suggest taking out the highlighting part on Days 0-3 • Suggest adding more interactive parts in the intervention (i.e., highlighting and generating their own questions) • Suggest spending more time on building the skills of partner sharing • Suggest Spending more time on explaining the meaning of making a connection in the introduction sessions • Suggest reducing the duration of intervention to less than 20 minutes • Suggest expending the group size to six • Suggest shortening into lessons to 20-25 min • Suggest adding more complex non-fiction readings questions • Suggest adding more time on practicing two-part connections

Research Question 2: What is the effect of the intervention on student reading outcomes?

Data analyses are still under way to address this question. Below are preliminary results using data that were available for analyses at the time of this report.

Pretreatment differences. A one between-groups analysis of variance (ANOVA) revealed no statistically significant pre-treatment differences between intervention and control students on screening measures (Maze, MAP, or CBM oral reading); see Table 26.

Table 26. ***Pre-Treatment Means,*** **SDs, F-*statistics, and Effect Sizes Comparing Intervention and Control Groups***

	Intervention (n= 28)		Control (n = 31)				
Measures	Mean	*SD*	Mean	*SD*	$F(1, 57)$	p	ES
Screening							
Maze (correct choices)	11.71	2.88	12.16	2.02	0.48	0.489	-0.18
MAP (standard score)	187.00	9.27	188.65	8.21	0.52	0.473	-0.19
CBM Oral Reading (wrc)	114.86	16.91	113.71	17.07	0.07	0.797	0.07

Near transfer: Stories. At pretest, there were no statistically significant differences between intervention and control students on the prompting task or on oral reading. At mid-test (after 3-weeks of intervention using stories), there was a statistically significant difference in the number of correct answers after prompting, with intervention students answering more questions correctly than control students (d = .56). See Table 27.

Near transfer: Biographies. At mid-test, there were no statistically significant differences between intervention and control students on the prompting task or on oral reading, although effect sizes favoring the intervention group were moderate. At posttest (after 3-weeks of intervention using biographies), there were no reliable between-group difference on any measure. See Table 27.

Posttests: Maze & MOCCA. A one between-groups analysis of variance (ANOVA) revealed no statistically significant post-treatment differences between intervention and control students on posttest measures (Maze or MOCCA); see Table 28.

Data remaining. When coding is completed, the following data will be analyzed:

- Near transfer (stories-pre/mid and biographies-mid/post)
    - Recall task
    - Highlighting task
- Far transfer (stories and biographies-pre/post)
    - Recall task
    - Oral reading

Table 27. ***Near Transfer: Questions Answered Correctly Before and After Prompting, and Words Read Correctly in One Minute***

	Intervention (*n*= 28)		Control (*n* = 31)		*F*		
Measures	Mean	*SD*	Mean	*SD*	(1, 57)	*p*	ES
Near transfer: Story (pre)							
Prompting							
Correct answers (before prompting)	2.46	1.00	2.19	1.05	1.03	0.315	0.26
Correct answers (after prompting)	3.82	0.39	3.81	0.48	0.02	0.896	0.02
Number of prompts needed	2.75	2.17	3.13	1.94	0.50	0.482	-0.19
Oral reading	136.29	24.43	130.32	20.86	1.02	0.316	0.26
Near transfer: Story (mid)							
Prompting							
Correct answers (before prompting)	2.71	1.08	2.52	0.96	0.50	0.484	0.19
Correct answers (after prompting)	***3.96***	***0.19***	***3.74***	***0.51***	***4.66***	***0.035***	***0.56***
Number of prompts needed	2.54	2.25	2.90	2.21	0.40	0.530	-0.16
Oral reading	147.11	27.74	136.74	23.62	1.33*	0.253	0.40
Near transfer: Biography (mid)							
Prompting							
Correct answers (before prompting)	2.61	1.03	2.29	1.07	1.33	0.253	0.30
Correct answers (after prompting)	3.93	0.26	3.77	0.50	2.15	0.148	0.40
Number of prompts needed	2.57	2.43	3.68	2.57	2.87	0.096	-0.44
Oral reading	113.14	22.73	111.71	21.21	0.06	0.803	0.07
Near transfer: Biography (post)							
Prompting							
Correct answers (before prompting)	2.54	0.92	2.55	0.89	0.01	0.909	-0.01
Correct answers (after prompting)	3.96	0.19	3.94	0.25	0.40	0.531	0.08
Number of prompts needed	2.43	1.97	2.77	2.14	0.14	0.712	-0.17
Oral reading	119.61	23.71	116.68	21.13	0.25	0.618	0.13

Table 28. ***Posttest Comparison: Maze and MOCCA***

	Intervention (*n*= 28)		Control (*n* = 31)				
Measures	Mean	*SD*	Mean	*SD*	*F*(1, 57)	*p*	ES
Maze (correct choices)	12.68	4.06	13.10	3.91	0.16	0.689	-0.11
MOCCA							
Correct	10.96	4.41	10.26	4.37	0.38	0.540	0.16
Paraphrases	4.18	4.00	4.58	3.97	0.15	0.700	-0.10
Bridging Inferences	1.89	1.47	1.97	1.20	0.05	0.830	-0.05
Lateral Connections	1.39	1.47	1.45	1.36	0.03	0.874	-0.04
Percent correct of attempted	0.60	0.24	0.56	0.230	0.42	0.518	0.17

Research Question 3: **Do subgroups of struggling readers (paraphrasers and elaborators) respond differentially to intervention?**

All of the above analyses will be conducted with subtype of struggling reader included as a moderator variable, when Think-Aloud and Recall coding are completed.

Round 14 (Spring Trial Run)

The purpose of the Spring Trial Run was to determine whether the final two lessons (Days 19 and 20) designed to apply the In-the-Text Connections approach to a regular text (with no questions inserted) were operating as intended and feasible to implement.

Following the Spring Pilot, participating teachers were given scripts and materials to implement Days 19 and 20 with their students (including copies of the book, *The Paper Bag Princess,* by Robert Munsch). Teachers were given the option to implement these lessons, and asked to provide feedback if they did so. At least two teachers implemented the lessons (Part 1 and Part 2), and completed a feedback form (in Appendix T). An intervention specialist who did not implement the intervention observed one who did, and also completed the feedback form. Their feedback is summarized in Table 29. Additional written feedback indicated teachers had a generally positive view about Days 19 and 20, and thought additional practice with new texts might be beneficial for some students.

Table 29. ***Teacher Feedback on*** **The Paper Bag Princess.**

Item	Responses
Part 1: Students identified important sentences in the text (Objective 1).	All students (1 teacher) Some students (2 teachers)
Part 1: Students connected important sentences to other parts of the text (Objective 2).	All students (1 teacher) Some students (2 teachers)
Part 1: Students' connections made sense.	All connections made sense (1 teacher) Some connections made sense (2 teachers)
Part 1: Which objective was most difficult for students to achieve?	Objective 1 (2 teachers) Objective 2 (1 teacher)
Part 2: Students identified important sentences in the text (Objective 1).	All students (2 teachers) Some students (1 teacher)
Part 2: Students connected important sentences to other parts of the text (Objective 2).	All students (2 teachers) Some students (1 teacher)
Part 2: Students' connections made sense.	All connections made sense (3 teachers)
Part 2: Which objective was most difficult for students to achieve?	Objective 1 (1 teacher) Objective 2 (1 teacher) Same difficulty (1 teacher)
Will students be able to generalize making connections to other texts without support?	All students (2 teachers) Some students (1 teacher)
Are these lessons *necessary* for generalization to occur?	Yes (2 teachers) No (1 teacher)
Are these lessons *sufficient* for generalization to occur?	Yes (1 teacher) No (1 teacher) Don't know (1 teacher)
Was the teacher guide helpful?	Yes (3 teachers)

C. Opportunities for Training and Professional Development

Opportunities for training/professional development (PD) to GRAs/URAs and teachers/specialists are describe in Table 30. PD opportunities include Knowledge Building activities, in which we provided opportunities for research staff and teachers to learn about important aspects of the grant and the intervention, Technical Training, in which we provided research staff with training to administer tests and intervention (much of which involves relevant knowledge and skills that are generalizable to their educational careers), and Mentoring (in which project faculty work with individual graduate students to develop their own research questions and analyses using project data).

Table 30. ***Professional Development Opportunities***

Type of PD	Content (Provided By)	Provided To
Knowledge Building	Overview/rationale for grant and intervention, theoretical and empirical underpinnings (McMaster, Janda) Implementing intervention (McMaster, Janda, Leinen, Lam)	Research Team (in Lab Meetings), Teachers (in Workshop)
Technical Training	Human subjects and responsible conduct of research training (University of MN) Administering and scoring CBM maze and oral reading (Pinto) Administering think-aloud (Janda & Pinto) Coding think-aloud (Janda) Administering and coding recall task (McMaster, Janda, Leinen)	GRAs and URAs
Mentoring	Participating in applied research (included writing a reflection on the experience) Writing research reports Developing independent research questions and data analysis (McMaster, Janda)	URAs GRAs GRAs

D. Results Disseminated to Communities of Interest

Results of the project have been shared with the following communities of interest (this list is cumulative for Years 1-3):

Community	Dates	Presentation
Participating teachers	May, 2012-14 January, 2014	Met with teachers in each school to share results and plan next steps Provided intervention workshop to participating teachers
Participating districts	August 2012 & 13	Provided annual report to participating districts
Minnesota reading practitioners	August, 2013; August, 2014	Presented workshop to reading practitioners attending the Minnesota Center for Reading Research Summer Workshop
UMN Special Education Program	April 4, 2012	Presented project to special education faculty and students
UMN College of Education	March 20, 2012; March 25, 2014	Presented posters at annual CEHD Research Day
National special education &	February, 2013	Presented research at the 2013 Pacific Coast Research Conference, San Diego, CA
education research community	March, 2014	Presented research at the 2014 SREE conference
National special education community	April, 2013 April, 2014	Presented research at the 2013 and 2014 Annual Convention of the Council for Exceptional Children

We also plan to share an abbreviated version of the Year 3 report with our two participating districts, and will offer to meet with interested school personnel to present our final Year 3 findings next school year. Additional conference proposals for posters and presentations are also underway.

E. Accomplishing Goals in the Next Reporting Period

In the next reporting period, we will complete Year 3 analyses; finalize revisions to the intervention, teacher workshop, and fidelity forms; and prepare presentations and papers for dissemination. We have requested, and have been granted, a no cost extension to complete the following:

Revision of intervention materials. During our Year 3 pilot study, we solicited feedback from teachers who implemented the intervention, along with intervention specialists, on the usability of our materials. I am requesting additional time to be able to thoroughly and systematically review their feedback and make final revisions to the materials, such that we have

a complete set of fully revised materials as a final product of this project. Revision to materials will be completed by **December 31, 2014.**

Year 3 analysis and writing. Preliminary analyses are underway, but due to the labor-intensiveness of coding our primary outcome variable (verbal recalls of text), and the need to wait until June to obtain district data (student demographics and achievement test data), a full analysis of all of our outcomes will be somewhat delayed this summer. We expect to complete preliminary analyses by **August 31, 2014**, but will continue follow-up analyses through December, 2014. Final reports and other dissemination activities (e.g., presentations to participating districts) will be completed by **August 31, 2015**.

Revision of teacher training materials. We also solicited teacher feedback regarding our teacher workshop. Teachers wanted to see more real-life examples of the intervention being implemented. This year, one of our outstanding school-based implementers agreed to allow us to video-record her as she delivered the intervention. We would like to edit the video and incorporate this into the teacher training in response to the feedback. Initial revisions will be made by **November, 2014** and training provided to participating teachers. We will solicit additional feedback, and final revisions to teacher training materials will be completed by **August 31, 2015**.

Revision of fidelity observation form. Finally, part of the Year 3 analysis will include examining which aspects of the intervention appear most critical for student success. Once we confirm the most critical components of the intervention, we will revise the fidelity observation form to reflect these critical components. Revision to the fidelity form will be completed by **August 31, 2015**.

II. PRODUCTS

A. Publications and Presentations

Presentations (cumulative for Years 1-3)

McMaster, K., Pinto, V., Janda, B., van den Broek, P., & Espin, C. (March, 2012). Making the right connections to improve the comprehension of struggling readers. Poster presented at the Annual University of Minnesota CEHD Research Day, Minneapolis, MN.

McMaster, K., Janda, B., & Pinto, V. (April, 2012). Under the big top: Balancing theory, evidence, and practice to develop a reading intervention. Presented to the monthly University of Minnesota Special Education Program Brown Bag Colloquium, Minneapolis, MN.

McMaster, K. L., Pinto, V., van den Broek, P., Espin, C., & Janda, B. (2013, February). Making the right connections: Improving the comprehension of struggling readers. Presented at the 2013 Pacific Coast Research Conference, San Diego, CA.

McMaster, K. L., Espin, C., van den Broek, P., Janda, B., Pinto, V., & Jung, P. G. (2013, April). Making the right connections: Improving the comprehension of struggling readers. Presented at the 2013 Annual Convention of the Council for Exceptional Children.

McMaster, K. L., Pinto, V., & Janda, B. (2013, August). Making In-the-Text Connections to understand stories and informational text. Minnesota Center for Reading Research Summer Literacy Workshop, St. Paul, MN.

McMaster, K. L., van den Broek, P., Espin, C., Janda, B., & Pinto, V. (2014, March). Making In-the-Text Connections to support struggling readers' comprehension. Presented at the 2014 Society for Research in Educational Effectiveness Conference, Washington, DC.

McMaster, K.L., Janda, B., Brodeur, K., Holmberg, N., & Jung, P. (2014, March). Making the Right Connections to improve the comprehension of struggling readers. Poster presented at the Annual University of Minnesota CEHD Research Day, Minneapolis, MN.

McMaster, K. L., Janda, B., Leinen, A. B., Hsu, H., Espin, C. A., & van den Broek, P. (2014, April). You're on the right track! Giving feedback during a reading comprehension intervention. Presented at the 2014 Annual Convention of the Council for Exceptional Children, Philadelphia, PA.

Publications

McMaster, K. L., van den Broek, P., Espin, C. A., Pinto, V., Janda, B., Lam, E. A., Hsu, H., Jung, P., Leinen, A. B., & van Boekel, M. (in press). Developing a reading comprehension intervention: Translating cognitive theory to educational practice. *Contemporary Educational Psychology.*

McMaster**,** K. L., Espin, C. A., & van den Broek, P. (2014). Making connections: Linking cognitive science and intervention research to improve comprehension of struggling readers. Invited article for *Learning Disabilities Research & Practice. 29*, 17-24.

Website
http://www.cehd.umn.edu/EdPsych/readinglab/

B. Other Products

Measures of fit, feasibility, and promise of the intervention (referenced as Appendices A-T in the report) have been developed and are attached.

III. PARTICIPANTS AND OTHER COLLABORATING ORGANIZATIONS

A. What individuals have worked on the project?

Name	Project Role	Person mo.	Contribution	Funding Support	Foreign Country
Kristen McMaster	Principal Investigator	3	Oversees all aspects of project	IES	NA
Paul van den Broek	Co-PI	2.4	Provides leadership in Discourse aspects of project	IES	Leiden University, The Netherlands
Christine Espin	Co-PI	1.2	Provides leadership in Intervention aspects of project	IES	Leiden University, The Netherlands
Sarah Carlson	Consultant	NA	Consulted regarding use of MOCCA for spring pilot	IES	NA
Viveca Pinto	Project Coordinator	3	Coordinated intervention activities	IES	NA
Bonita Janda	Project Coordinator	6	Coordinates discourse activities	IES	NA
Amy Bethel Leinen	GRA/Project Coordinator	3	Coordinated and implemented data collection & analysis, intervention development & implementation, report writing	IES	NA
Pyung-Gang Jung	GRA	3	Data collection & analysis, intervention development & implementation, report writing	IES	NA
Hui-Chen Hsu	GRA	6	Data collection & analysis, intervention development & implementation, report writing	IES	NA
Elizabeth Oberg Lam	GRA	3	Data collection & analysis, intervention development & implementation, report writing	IES	NA
Jaehyun Shin	GRA	3	Data collection & analysis, intervention development & implementation	IES	NA
Kendra Rimmereid	GRA	3	Data collection & analysis, intervention development & implementation	IES	NA
Katherine Brodeur	GRA	3	Data collection & analysis, intervention development & implementation	IES	NA
Martin van Boekel	GRA	3	Data collection & analysis, intervention development & implementation, report writing	IES	NA
Neil Holmberg	GRA	1.5	Data collection	PI non-sponsored funds	NA

In addition, Undergraduate Research Assistants (URAs; junior or senior Psychology majors) worked on the project as volunteers or for course credit for varying numbers of hours.

- Sukaina Ali (fall and spring)
- Anne Bonanno (fall and spring)
- Mackenzie Hoffman (fall and spring)
- Neil Holmberg (fall only)

- Shinsig Lee (fall and spring)
- Allison Morrow (spring only)
- Mari Nyhus (fall and spring)
- Allison Wilcox (fall and spring)
- Nicole Woodall (fall only)

What other organizations have been involved as partners?
Bloomington Public Schools and Minneapolis Public Schools served as the district sites for all school-based activities described in this report.

Have other collaborators or contacts been involved? No.

IV. IMPACT

A. Impact on the development of the principal disciplines of the project

The current project has the potential to impact the principal disciplines (cognitive science, reading research) in the following ways:

(1) We are shedding light on how cognitive models of reading can be translated into reading intervention for struggling readers. Particularly, we are (a) examining whether cognitive theories about the importance of causal connections can be translated into a questioning intervention, and (b) determining whether subgroups of struggling readers (identified using cognitive processing methodologies) are differentially responsive to different approaches to reading intervention. In turn, results may help refine cognitive models of reading. For example, our findings suggest that students who struggle with reading *are able* to make text-based, causal connections; however, they do not necessarily do so unless they are provided with specific questions and scaffolded feedback.

(2) Findings may inform the field about the most efficient and promising way to individualize intervention in a small-group context.

(3) We are determining critical features that increase the feasibility of implementing reading comprehension intervention in school settings, including group size (no fewer than 3-4 students), frequency and duration (3 times per week for 20-30 min), interventionist (teacher or tutor with substantial training and support), highly scripted and well organized materials packaged for immediate use, development of texts and questions that fit both the intervention and schools' instructional programming.

B. Impact on other disciplines

None at this time.

C. Impact on the development of human resources

This project has provided opportunities for graduate and undergraduate research assistants and teachers to learn about cognitive models of reading, intervention design, research design, and to gain experience implementing a variety of assessments and intervention protocols.

One of the undergraduate psychology students who volunteered as a research assistant and earned academic credit in Years 1 and 2, subsequently applied and was accepted into the Master's in Education program in special education and became a GRA in Year 3.

D. Impact on resources that form infrastructure

None at this time.

E. Impact on technology transfer

None at this time.

F. Impact on society

None at this time.

G. Dollar amount spent in foreign countries

$123,150.35 in the current reporting period (subcontract with Leiden University, in The Netherlands).

V. CHANGES/PROBLEMS

There have been no changes to the activities outlined in the Performance Agreement (as revised on 8/17/12), and no problems in the current reporting period. No changes or problems are anticipated for the upcoming reporting period.

VI. BUDGETARY INFORMATION

Budget Justification

A. Key Personnel

Kristen McMaster, Principal Investigator, (2.25 Cal. Months AY/2 Cal. Months summer)
Dr. McMaster has devoted 25% FTE (2.25 cal months AY and 2 cal months summer) to this project. She oversees all major grant activities, including management of the budget; supervision of Site and Intervention coordinators, GRAs, and data collectors; development and revisions of intervention materials and procedures; conducting training of interventionists and teachers; data collection from various sources (experts, teachers, pilot studies); designing pilot studies; data analysis and interpretation. She also prepares reports required by the agency.

B. Other Personnel

Bonita Janda, Discourse Coordinator (6.5 cal months over 12 months)
Ms. Janda She provides technical assistance for this project under the direct supervision of the PI. Specifically, she supervises GRAs and URAs; participates in development and revisions of intervention materials and procedures; coordinates data collection efforts with experts; coordinates scheduling for screening and other assessments; coordinates scheduling for interventions conducted during trial runs and pilot studies; oversees data transcription, coding, and entry; assists with data analysis and interpretation; and assists with writing reports.

Viveca Pinto Intervention Coordinator (4.5 cal months over 12 months)
Dr. Pinto devoted 50% FTE (6 cal months over 12 months) to this project through February 11, 2014. She left the project at that time to care for her newborn child. Her responsibilities were assumed by a GRA, whose time was increased from 25% to 50% FTE. The intervention coordinator coordinates development and revisions of intervention materials and procedures; coordinates data collection efforts with teachers, including facilitating focus groups; coordinates and participates in training of interventionists and teachers; conducts interventions and observations during intervention trial runs and pilot studies; assists with data analysis and interpretation; and assists with writing reports.

Graduate Research Assistants (26 cal months) assist with development and revisions of intervention materials and procedures; administer and score screening measures and other assessments; serve as interventionists; code Think-Aloud protocols and data obtained from questionnaires, surveys, and observations; attend meetings as needed; and assist with data entry and analysis.

C. Travel

Conference Travel and Mileage (budgeted = $5,000 per year, Year 2 expense = $7354.00)
Included travel to two conferences to present Year 1 findings, and mileage for project staff to travel to schools.

D. Other Direct Expenses

Materials and Supplies (budgeted=$4,300 Yr. 3, actual=$579.)
These funds were used for supplies necessary for project implementation (e.g., intervention binders and folders). Supply expenses have not been as high as anticipated, as it was not necessary to purchase assessments and software that were available through other resources in the department.

Consulting (budgeted = $6,000/year, actual Yr. 3 = 0)
Given that Year 3 was a Pilot year and we had gained input needed from consultants in Years 1-2, we did not need to use consultant funds for Year 3.

Subaward (budgeted=$107,695 Yr. 2; actual = $123,150.35)
Drs. Paul van de Broek and Christine Espin of Leiden University serve as co-investigators on the project. The slightly higher-than-budgeted amount reflected here is due to the fact that the timing of activities (and thus, invoicing) has been slightly different than was reflected in the original budget (for example, last year's Subaward expense was $94,215 and the budgeted amount was $103,419). We expect the final subaward to approximate the actual budgeted amount.

Teacher Stipends and Substitute Costs (budgeted = $24,320 Yr. 2, actual = $9,307)
Funds were used to pay teacher stipends and reimburse substitutes needed to release teachers for the teacher workshop. Substitute costs were not as high as anticipated because only a subset of teachers needed to attend the teacher workshop.

Other expenses Copying, Background Checks, Student Incentives, Room Rental (budget = $6,025 Yr. 3, actual = $3120) Funds were used for copying background checks (for GRAs to work in schools), student incentives (to purchase gel pens and stickers to be used as incentives for intervention) and room rental (for teacher workshop). Costs were not as high as expected as we were able to find discounted items for student incentives and minimize copying.

E. Indirect Costs

The University of Minnesota applies a 34% Indirect Cost rate to all direct costs minus graduate research assistant fringe and the amount of the subcontract over $25,000. The on-campus indirect cost rate is being applied given that more than 50% of personnel on this project are housed on-campus.

Changes to the Budget
There were no changes to the budget that resulted from modifications to project activities.
There were no changes to the budget that affected our ability to accomplish project activities as described in the performance agreement

We do not anticipate that funds will be carried over to the next project period. The balance as of this report was $156,876. As of July 24, 2014, the balance is $65,674. This amount should be spent down by August 31, 2015 during our no cost extension period. There are no anticipated changes in the budget for the next project period.